ISO ERC™ Suite Product Guide

October 2019 Edition

Edition – October 2019

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Preface

There are four guides that make up the product documentation for ISO Electronic Rating Content™ (ERC™). This document, the *ISO ERC™ Suite Product Guide*, explains each of the components of ERC – the rating content, algorithms, release notes, readme file, and sample test case. The other three ERC guides are listed and briefly described below.

ISO ERC™ Maintenance Guide

This guide outlines various scenarios that involve rating content changes that may occur from one release to another and how these changes would be implemented within the ERC product. It also describes the Automated Maintenance Feed (AMF), a single XML file that includes the most recent changes to ISO rating data such as loss costs and rating algorithms to facilitate the import of the changes only into a rating system.

ISO Rating Content XML Guide

This guide outlines the structure of the XML for the rating content.

ISO ERC™ On-line Help Guide

This guide explains how to work with the ERC product from the ISOnet user interface.

Overview

ISO Electronic Rating Content (ERC) provides a package of rating information that represents the ISO Commercial Lines Manual (CLM), integrating all the applicable countrywide rules, state exceptions, loss costs, and form attachment logic for each line of business and state that are effective at a certain point in time. This information is available in Excel and XML formats that lay out the rating data (e.g., limits, deductibles, advisory loss costs, and classification codes) and algorithms (e.g., form attachment logic, statistical code assignment, rating messages, and premium development and rating procedures) for working with the rating data so that the information can be uploaded by carriers and vendors to integrate with any rating and policy administration system.

Lines of Business and States Supported

ERC is available for the following lines of business in all states and jurisdictions supported by ISO:

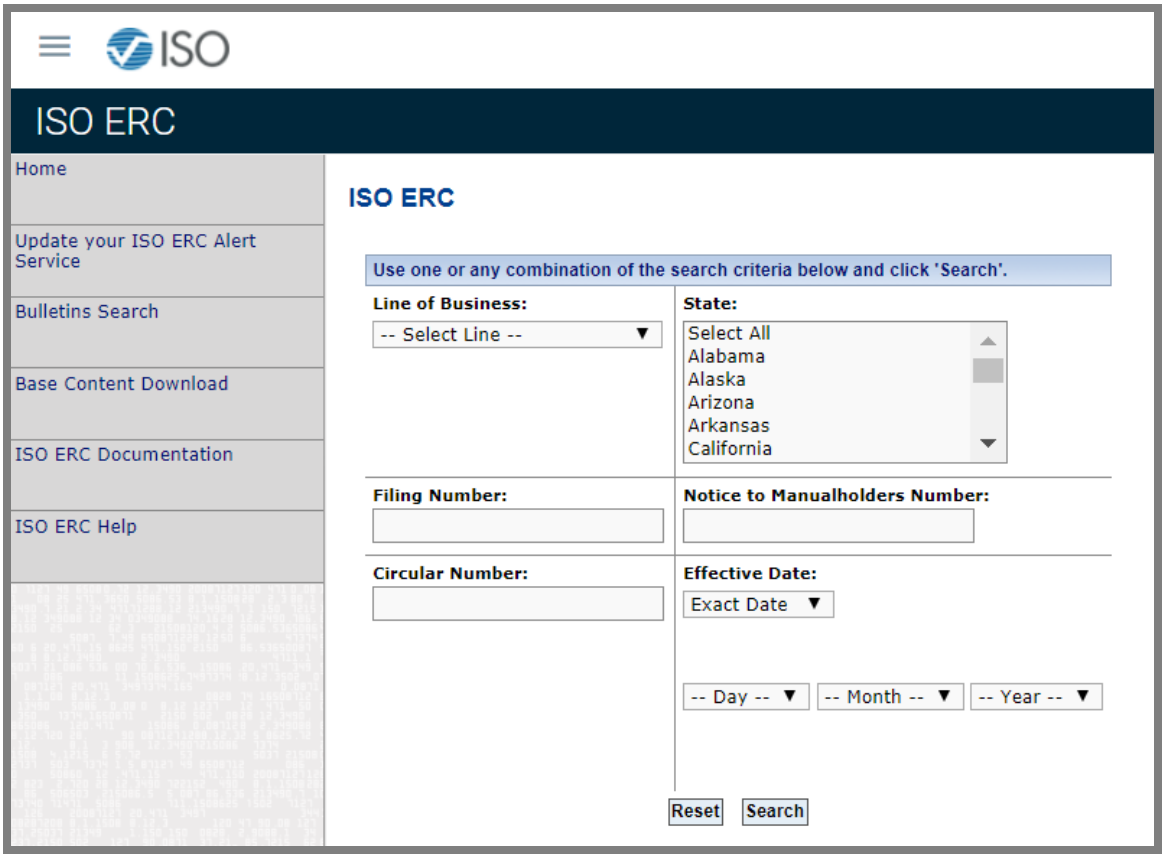
- Businessowners
- Commercial Automobile
- Commercial Inland Marine (Filed and Non-Filed)
- Commercial Package Policy
- Commercial Property
- Commercial Umbrella
- Crime
- Cyber
- General Liability

ERC also supports the following non-ISO line(s) of business:

- Workers Compensation

Where to Access ERC Information

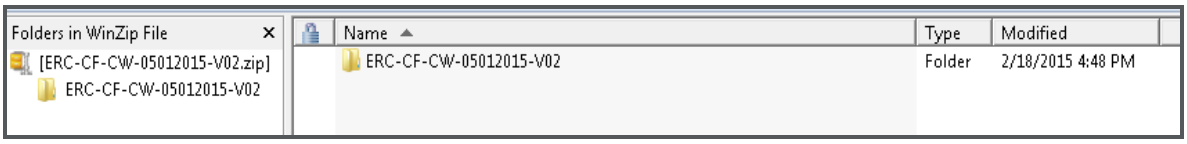
Access to ERC information, including the rating content and algorithm files, as well as update and support information, is available through the ISOnet user interface. See the screenshot below of the ERC home page which is the starting point for searching, selecting, and downloading ERC information.



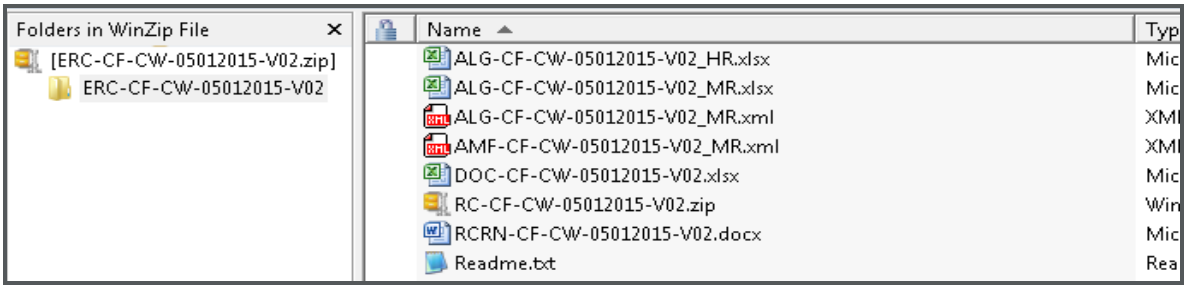
For more information about how to access ERC via ISOnet and set up e-mail alerts about product updates, please refer to the *ISO ERC On-Line Help Guide* on the ERC home page under the “ISO ERC Help” link.

Components of ERC Package

Once an ERC or “Ratebook Content” package is opened from the search results page, a WinZip file containing an ERC folder will be displayed. This is where all the files are stored that make up the specific Ratebook Content package selected. See the example below.



Once the ERC folder is opened, all the files contained in that folder will be displayed. See the screenshot below followed by a description of the file name format and each type of file.



ERC File Name Format

ERC files use the format TTTT-LL-XX-MMDDYYYY-VVV:

- TTTT – Identifies the type of file (e.g., ALG, AMF, DOC, RC, RCRN, and TC).
- LL – Identifies the two-character alpha code for a line of business (e.g., CF for Commercial Property).
- XX – Identifies the two-character alpha code for a state or “CW” for countrywide.
- MM/DD/YYYY – Identifies the two-digit month and day followed by the four-digit year representing the effective date of the rating content. Note: Effective date is driven by the effective date of circulars that approve modifications to the ISO CLM.
- VVV – Identifies the version number of the content. Occasionally an ERC package that has already been released to ISOnet needs to be updated to address a correction or enhancement. Whenever this occurs, the ERC package is re-released with a new version number.

Documentation File

The documentation or “DOC” file provides information about specific ISO CLM rules and plans that are not supported within ERC or that are “Refer to Company”. This file also provides special considerations related to the line of business, such as rule number guidelines and stat assignment. It is important to review this file before beginning work with the rating content and algorithms to make sure you know what is Refer to Company and not supported in the specific ERC package.

The DOC file is in an Excel format and contains several different worksheets or tabs. See the list of tabs below followed by a brief description and illustration of each.

- Table of Contents
- Refer to Company
- Not Supported
- Rule Number Guideline
- Stat Assignment
- Special Considerations

Table of Contents Tab

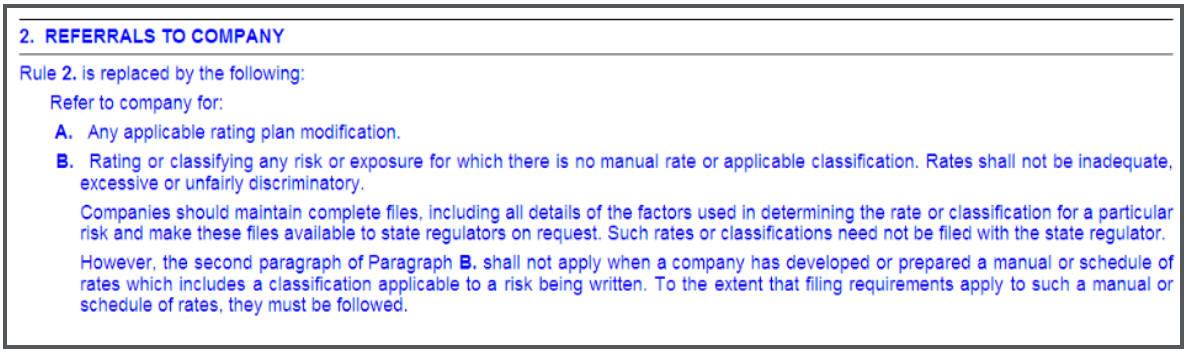
The Table of Contents tab lists the sections of the DOC file including links to each section. See the example below. Note: The sections included in a DOC file will vary, depending on the ERC package. However, most DOC files include Refer to Company and “Not Supported” tabs.

A	B	C
Algorithm	**Comments**	
Refer to Company		
Not Supported	Updated	
Rule Number Guideline		
Stat Assignment		
Special Considerations		

Refer to Company Rules Tab

The ISO CLM contains many Refer to Company rules. These are rules that instruct the user to check with their company for information regarding the implementation of rating

content which may be dictated or based on a company filing. For example, Rule 2. REFERRALS TO COMPANY in the State Insurance Manual (SIM) for Commercial Property Alabama (SIM-CF-AL-05/01/2015) states the following:



2. REFERRALS TO COMPANY

Rule **2.** is replaced by the following:

Refer to company for:

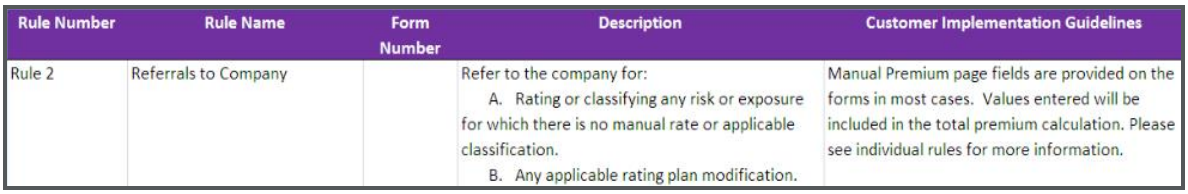
A. Any applicable rating plan modification.

B. Rating or classifying any risk or exposure for which there is no manual rate or applicable classification. Rates shall not be inadequate, excessive or unfairly discriminatory.

Companies should maintain complete files, including all details of the factors used in determining the rate or classification for a particular risk and make these files available to state regulators on request. Such rates or classifications need not be filed with the state regulator.

However, the second paragraph of Paragraph **B.** shall not apply when a company has developed or prepared a manual or schedule of rates which includes a classification applicable to a risk being written. To the extent that filing requirements apply to such a manual or schedule of rates, they must be followed.

To the degree that Refer to Company rules are supported in ERC, information is provided in the "Customer Implementation Guidelines" column of the Refer to Company tab. See the example below.



Rule Number	Rule Name	Form Number	Description	Customer Implementation Guidelines
Rule 2	Referrals to Company		Refer to the company for: A. Rating or classifying any risk or exposure for which there is no manual rate or applicable classification. B. Any applicable rating plan modification.	Manual Premium page fields are provided on the forms in most cases. Values entered will be included in the total premium calculation. Please see individual rules for more information.

Not Supported Rules Tab

"Not Supported" rules are rules that are not supported in ERC. For example, Rule 85.N. Transition Rule for Changes in Class Rating Eligibility – Light Manufacturing Classes and Hotels/Motels in the State Insurance Manual (SIM) for Commercial Property in Alabama (SIM-CF-AL-05/01/2015) states the following:



85. BASIC GROUP I CLASS RATES

The following is added to Rule **85.**:

N. Transition Rule For Changes In Class Rating Eligibility - Light Manufacturing Classes And Hotels/Motels

1. This transition rule is applicable to the first and second renewals of coverage on property that was previously specifi

0533	Baking on premises, no delivery to outlets
0745	Motels and hotels without cooking - Up to 10 units
0746	Motels and hotels without cooking - 11-30 units
0747	Motels and hotels without cooking - Over 30 units
2200	Baking on premises, delivery to

For this rule, the ERC Customer Implementation Guideline refers the user to the company for information about how to implement this rule. See the example below.

Rule Number	Rule Name	Form Numbe	Description	Customer Implementation Guidelines	C
Rule 85.N	Transition Rule For Changes in Class Rating Eligibility - the state exception ratebook does not include the transition rule.		Transition Rule For Changes in Class Rating Eligibility has not been implemented	The company will be responsible for developing and implementing all transition rules.	

Rule Number Guideline Tab

The “Rule Number Guideline” tab provides information about how ISO rule references are formatted and documented within ERC. See the examples below:

	A	B	C
1	Database Table Association	Bureau Rule Number (Example)	Comment (E
2	The Bureau Rule Number column will not include the word "Rule".	28 Bldg	
3	If a database table is referenced by more than one Rule, all applicable Rules will be listed, separated by semicolons	38.H; 38.J PrsnlProp	
4	The Bureau Rule Number will be followed by the Level: Policy; Location; Bldg for Building; PrsnlProp for Personal Property; BI for Business Income; SpCl for Special Class and SpCl BI for Special Class Business Income.	38.E Location	

Stat Assignment Tab

The “Stat Assignment” tab provides information about how stat codes are assigned within ERC. See the example below. Note: The Page Number column refers to the page in the ISO Statistical Manual that addresses the specific stat code.

Code Assignment	Page Number	Customer Implementation Guidelines
Automobiles Furnished for Regular Use: Private Passenger	CA-LI-39.2; CA-PH-35.2	Rate table PrivatePassengerAutosFurnishedForRegularUseStatCode is added and AutosFurnishedForRegularUse page field is added on Private Passenger Detail page

Special Considerations Tab

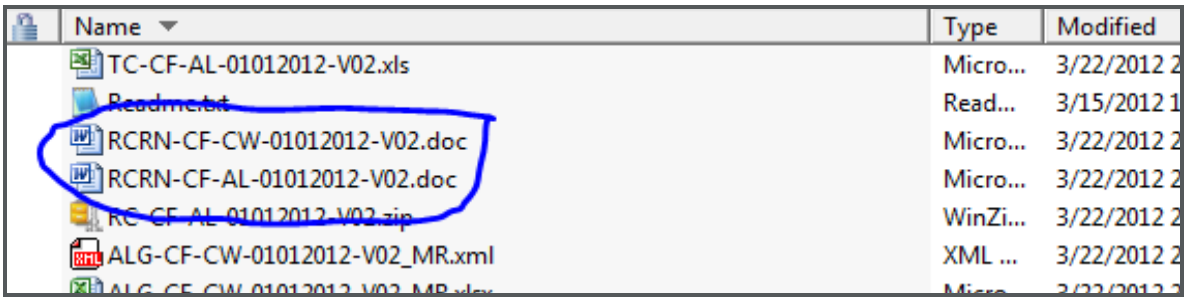
The “Special Considerations” tab provides additional information regarding the line of business that may be unique to the line and important to note during the implementation. See the example below.

Content Special Considerations	
Other Public Buses is rated as Other Buses	Class codes classified under "Other Public Buses" will use the same loss cost values as "Other Buses". This will be reflected in the rate table PublicTransportationGroup

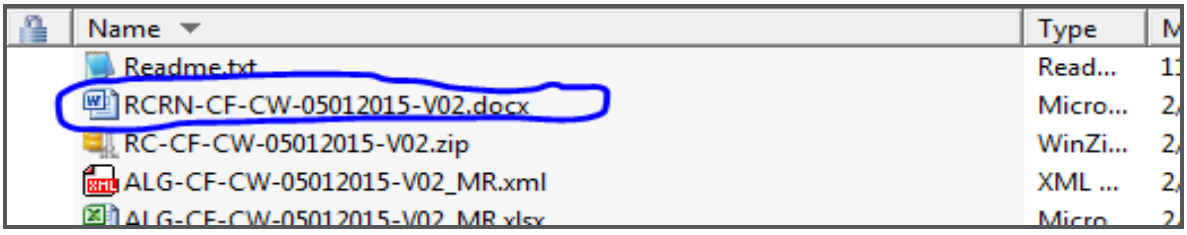
Rating Content Release Notes (RCRN) File

The Rating Content Release Notes (RCRN) file is a Word document that describes in detail the changes made within a specific ERC package as of the effective date identified in the RCRN file name. The types of changes that can be made to an ERC package include approved changes announced in ISO Circulars, as well as corrections, enhancements, and maintenance items. It is important to review release notes to be aware of changes made to the rating content and algorithms for implementation, as well as maintenance purposes.

Depending on the effective date of the ERC package, there can be one or two RCRN files. ERC packages released prior to June 2014 contain state and countrywide files bundled together in a single ERC package and two RCRN document files, one countrywide version and one state version. See the example below.



ERC packages released after June 2014 split state and countrywide files into separate ERC packages each with its own RCRN file. Countrywide ERC packages include a countrywide version of the release notes; state ERC packages include a state-specific version. See the example below.



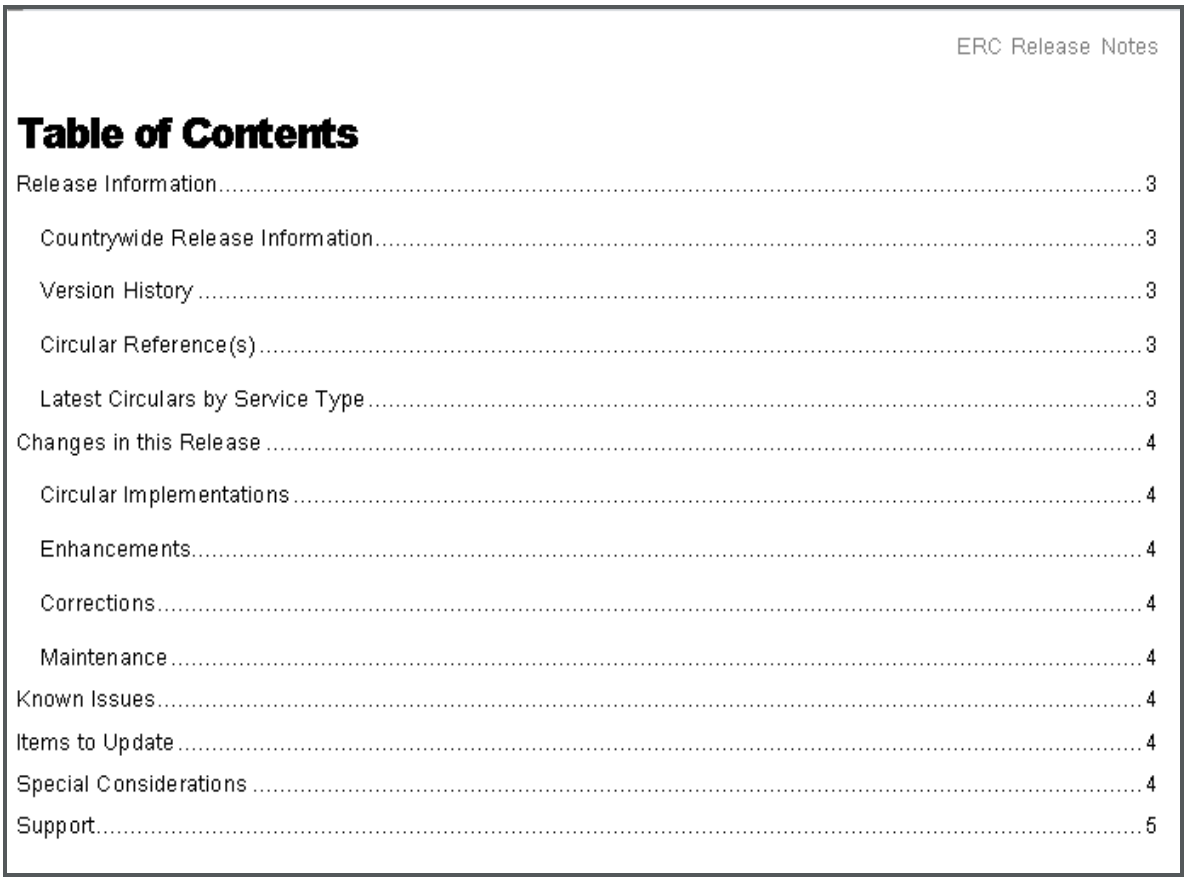
Also, as of February 2016, changes to values made in the rate tables and domain tables found in the Rating Content ("RC") file and changes to PCH and workflow changes found in the Algorithm ("ALG") file are available in an XML format. This information is stored in a separate maintenance file called the "AMF" (Automated Maintenance Feed)

file. Please refer to the *ISO ERC Maintenance Guide* for more information about this XML file.

See a list of release notes sections below followed by a description and illustration of each:

- Table of Contents
- Release Information
  - Countrywide Release Information
  - Version History
  - Circular Reference(s)
  - Latest Circulars by Service Type
- Changes in this Release
  - Circular Implementations
  - Enhancements
  - Corrections
  - Maintenance
- Known Issues
- Items to Update
- Special Considerations
- Support

Table of Contents – The table of contents summarizes the sections of the release notes. Below is a screenshot of the table of contents page for the release notes contained in the Commercial Property countrywide package, RC-CF-CW-05012015-V02.



ERC Release Notes

Table of Contents

Release Information 3

Countrywide Release Information 3

Version History 3

Circular Reference(s) 3

Latest Circulars by Service Type 3

Changes in this Release 4

Circular Implementations 4

Enhancements 4

Corrections 4

Maintenance 4

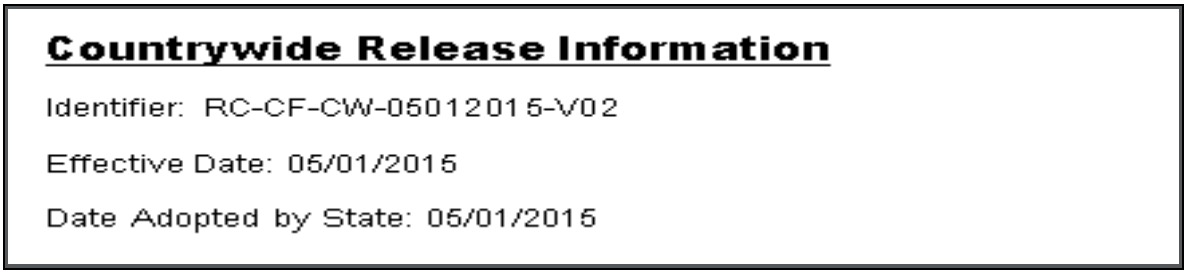
Known Issues 4

Items to Update 4

Special Considerations 4

Support 5

Countrywide Release Information - This section identifies the name and effective date of the ERC package being released and also provides other information pertinent to the release including version history, circular references, and the latest circulars by service type. See the example below.



Countrywide Release Information

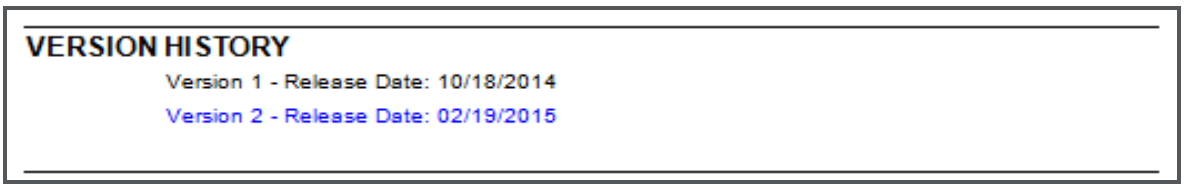
Identifier: RC-CF-CW-05012015-V02

Effective Date: 05/01/2015

Date Adopted by State: 05/01/2015

As the example shows, this is a second version of the countrywide content which is denoted by “V02” in the file name. Whenever a file is re-released with changes to the content, the file is assigned a new version number. All the items contained within the CF-CW-05012015-V01 release also apply to CF-CW-05012015-V02, as well as the new items added. Note: Throughout the release notes document, the text highlighted in blue font identifies information about the new items added to a release.

Version History – This section identifies the date each version of the content was released to ISOnet. See the example below. In this case, CF-CW-05012015-V01 was released to ISOnet on 10/18/2014 and CF-CW-05012015-V02 was released on 2/19/2015.



Circular Reference(s) - This section summarizes information about the ISO Circulars that are being implemented in the ERC package including the ISO Circular name, number, date released, and type, as well as the corresponding Filing ID or Filing Reference number and Notice to Manualholders (NTM) number. See the example below.



Latest Circulars by Service Type - This section identifies the most recent ISO Circulars that have been implemented by service (ISO Loss Costs, Rules, Forms, and Statistical Plan), as well as the effective date for each type of circular. See the example below.

Service Type	Circular Number	Circular Effective Date
FORMS	LI-CL-2015-106	05012016
RULES	LI-CF-2014-080	11012014
LOSS COST	LI-CF-2014-080	11012014
STATISTICAL PLAN	SP-CF-2014-002	05012015

Changes in this Release – This section describes the changes within the ERC package including all circular implementations, as well as corrections, enhancements, and maintenance items.

Circular Implementations - This section provides a detailed list of the changes associated with the ISO Circular that were implemented in the ERC package. Examples of changes include the addition, revision, or deletion of Tables, Columns, Pages, Page Fields, Domain Tables, and Rate Tables, as well as changes to algorithms. See the example below.

The following form edition date(s) were revised:

Name	Previous Edition Date	New Edition Date
Maryland Changes	IL 02 07 04 11	IL 02 07 01 15

Enhancements - This section provides detailed information about any enhancements implemented in the ERC package. See the example below.

ENHANCEMENTS

1. **Description:**

   The intermediary premiums of Pre TRIA Premium and Post TRIA Premium should be determined before the final calculations begin to make it clearer in the ALG-CF-CW-05012015-V01. Other code in Terrorism was streamlined, particularly the calculation for IL 09 85. There was no rating impact to these changes.

   Enhancement:

   In the ALG-CF-CW-05012015-V01 please refer to:

   'PreTRIAPremium'

   'PostTRIAPremium'

   were updated in the Building Common Rating tab.

Corrections - This section provides detailed information about any corrections made to the ERC package including a brief description of the problem and the solution. See the example below.

Corrections

1. Description:

In Special Class Detail page, the page field InflationGuard was using the wrong domain table InflationGuardPctPrsnlProp instead of using the Countrywide domain table InflationGuardPct. The page field was inherited.

Correction Detail:

The following page field(s) were revised:

Page Field Name	Page Name
InflationGuard	Special Class Detail

Maintenance – This section provides detailed information about any maintenance items included in the release. See the example below.

MAINTENANCE

1. On Location Detail page, page field SpecialRatingTerr had a default value that was no longer necessary. It was eliminated.

Known Issues - This section identifies any known issues being released in the ERC package.

Support - This section provides a link to the ERC Customer Support Portal where cases concerning questions or issues are created and submitted to the ERC Support Team.

Release Notes for State ERC Packages

All the sections described above for the countrywide RCRN document are contained in the state RCRN file, as well as a section for State Release Information. This section identifies the name and effective date of the state package. See the example below.

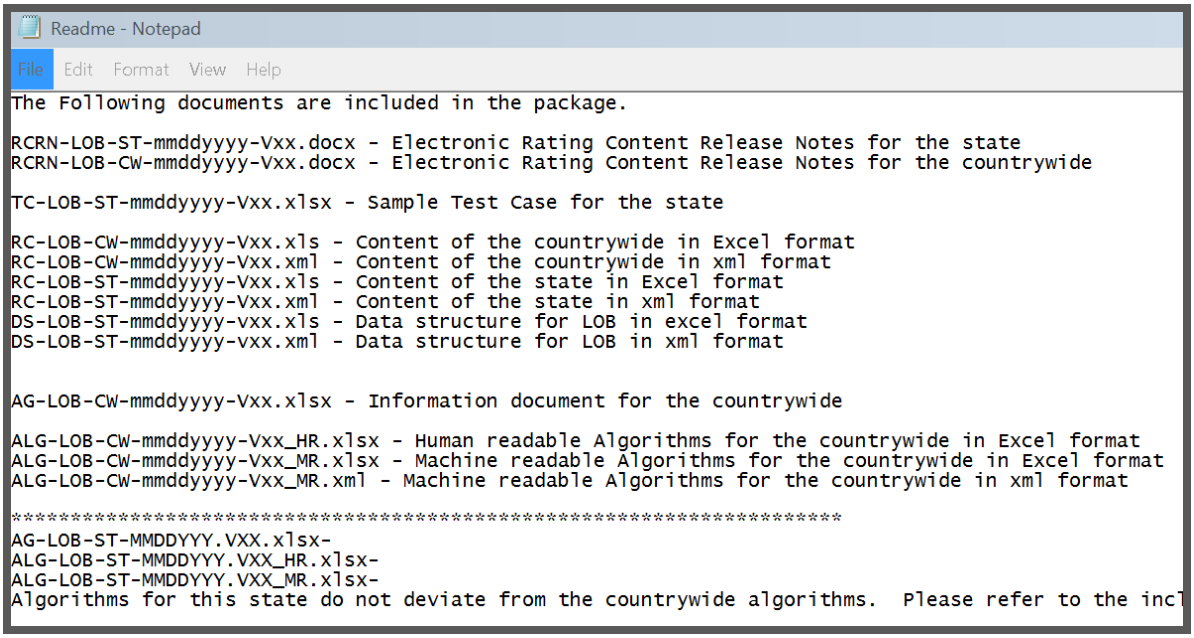
Maryland Release Information

Identifier: RC-CF-MD-0501 2016-V01

Effective Date: 05/01/2016

Readme Text File

For information purposes, a generic Readme Text file is distributed with every ERC package. This file contains a list and brief description of all the files distributed within an ERC package including countrywide, state, Excel, XML, and human readable (HR), and machine readable (MR) versions of the files. See the example below. This information may be used for audit purposes, i.e., every file identified in the Readme file should be included in the ERC package you are working with.

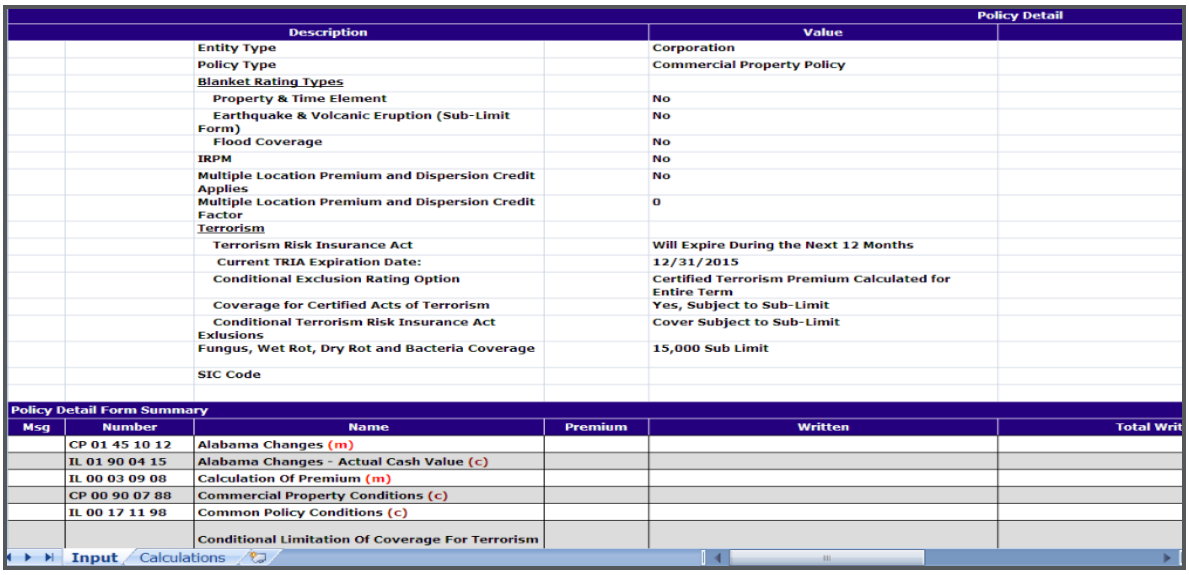


Test Case (TC) File

Every state ERC package includes a Test Case (TC) file that provides a sample test case. The sample test case is an Excel representation of a policy that outlines the data used at each risk level and the sub premiums and total premium generated from this data.

Below is a list and brief description of the worksheets or tabs within the TC file followed by an illustration.

- **Input -** Provides the data or values used in the test case policy.
- **Calculations -** Provides the premiums.



Policy Detail

Description	Value
Entity Type	Corporation
Policy Type	Commercial Property Policy
Blanket Rating Types	
Property & Time Element	No
Earthquake & Volcanic Eruption (Sub-Limit Form)	No
Flood Coverage	No
IRPM	No
Multiple Location Premium and Dispersion Credit Applies	No
Multiple Location Premium and Dispersion Credit Factor	0
Terrorism	
Terrorism Risk Insurance Act	Will Expire During the Next 12 Months
Current TRIA Expiration Date:	12/31/2015
Conditional Exclusion Rating Option	Certified Terrorism Premium Calculated for Entire Term
Coverage for Certified Acts of Terrorism	Yes, Subject to Sub-Limit
Conditional Terrorism Risk Insurance Act Exlusions	Cover Subject to Sub-Limit
Fungus, Wet Rot, Dry Rot and Bacteria Coverage	15,000 Sub Limit
SIC Code	

Policy Detail Form Summary

Msg	Number	Name	Premium	Written	Total Writ
	CP 01 45 10 12	Alabama Changes (m)			
	IL 01 90 04 15	Alabama Changes - Actual Cash Value (c)			
	IL 00 03 09 08	Calculation Of Premium (m)			
	CP 00 90 07 88	Commercial Property Conditions (c)			
	IL 00 17 11 98	Common Policy Conditions (c)			
		Conditional Limitation Of Coverage For Terrorism			

Input Calculations

Rating Content (RC) File

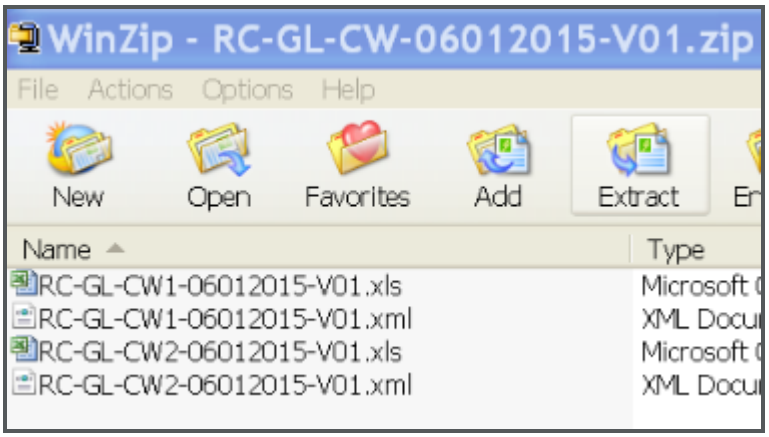
The Rating Content or “RC” file contains all the rating data (e.g., limits, deductibles, advisory loss costs, and classification codes) and requirements (e.g., domain tables) for a line of business. This information is provided in both Excel and XML formats. See the example of the RC files below for Commercial Property countrywide content effective 05-01-02015 (CF-CW-05012015-V02).



Name	Type	Modified	Size	Ratio	Packed
RC-CF-CW-05012015-V02.xls	Micro...	2/2/2015 10:47 AM	70,435,...	99%	877,404
RC-CF-CW-05012015-V02.xml	XML F...	2/2/2015 10:26 AM	16,717,...	95%	856,988

If you are involved in building an interface or tool to programmatically consume ERC into a rating system, you will most likely work with the XML format which provides a machine-readable version of the RC file; otherwise, you should work with the Excel format.

Except for Commercial Auto and General Liability, countrywide rating content is provided in two files, as shown above. However, for Commercial Auto and General Liability there are four files, two countrywide or “CW1” files and two “CW2” files. See the example below.



If you are working with the rating content for Commercial Auto or General Liability, in your initial analysis of the content, you will need to work with both the CW1 and CW2 files to get a complete picture of the line of business. Once you have completed your initial analysis, you will only need to work with the CW2 file for maintenance purposes.

While the countrywide RC file provides two files, the RC file for state content provides four files – two “RC” files and two “DS” files. See the example below.



Data Structure (DS) Files

The “DS” or data structure files provide an overview of the line of business, as of the effective date of the content (e.g., 05-01-2015). All the tables involved in building the structure of the line of business are in this file including active and obsolete tables at the countrywide and state levels. In addition, this file illustrates the relationship of these tables, i.e., the parent/child relationship.

The reason why the DS files are included in the state file and not the countrywide file is because countrywide files do not change once they are released whereas state files typically involve changes.

Below is a list and brief description of the tabs within the DS file.

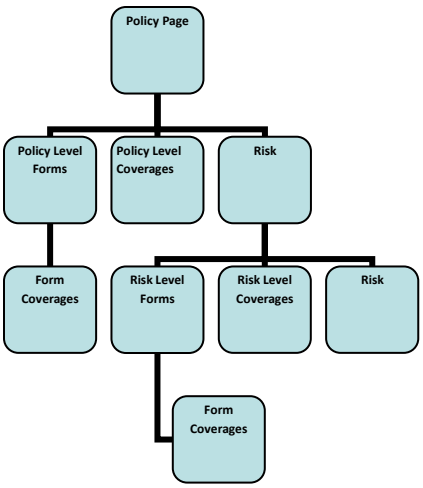
- Tables
- Column Details
- Ratebook Tables

DS File - Tables Tab

The Tables tab provides a list of all the tables that make up the line of business and the parent/child relationship these tables have to one another. See the example below followed by a brief description of each column.

	G	H	I	J	K	L
6	***Table***	***Parent***	***Type***	***Override***	***Description***	***Comment***
7	CommercialProperty		policy	0	Commercial Property	
8	CommercialPropertyAaTest	CommercialProperty	form	0	Ruleset Test Form	
9	CommercialPropertyAdditionalLocationsSpecialCoinsuranc	CommercialProperty	form	0	Additional Locations Special Coinsurance General Informat	Additional Locations Sp
10	CommercialPropertyAdditionalLocationsSpecialCoinsuranc	CommercialPropertyAdditionalLocationsSpecia	coverage	0	Additional Locations Special Coinsurance General Informat	
11	CommercialPropertyAdditionalLocationsSpecialCoinsuranc	CommercialPropertyAdditionalLocationsSpecia	coverage	0	Additional Locations Special Coinsurance General Informat	
12	CommercialPropertyAdditionalLocationsSpecialCoinsuranc	CommercialPropertyAdditionalLocationsSpecia	coverage	0	Additional Locations Special Coinsurance General Informat	
13	CommercialPropertyAdditionalLocationsSpecialCoinsuranc	CommercialPropertyAdditionalLocationsSpecia	coverage	0	Additional Locations Special Coinsurance General Informat	
14	CommercialPropertyAdditionalLocationsSpecialCoinsuranc	CommercialPropertyAdditionalLocationsSpecia	coverage	0	Additional Locations Special Coinsurance General Informat	
15	CommercialPropertyAlabamaChangesCancellation	CommercialProperty	form	0	Alabama Changes - Cancellation	
16	CommercialPropertyAlaskaChangesFungusWetRotAndDry	CommercialProperty	form	0	Alaska Changes Fungus Wet Rot And Dry Rot	
17	CommercialPropertyAlaskaExclActsBiolgclChmclTerrsmCa	CommercialProperty	form	0	Alaska - Exclusion of Acts of Biological or Chemical Terror	

- **Table** - Provides the name of the table.
- **Parent** - Represents the relationship the table has to other tables. In the example above, the “Commercial Property” table is the parent of the “Commercial Property Additional Locations Special Coinsurance General Information” table.
- **Description** - Provides a brief description of the table.
- **Type** - Identifies the type of table. There are five types. See diagram and description of each below.



Types of Tables:

1. **Policy** – The policy level represents the policy table in the rating content database. There can only be one policy per record; this is the starting point of the hierarchy of rating content. The following items can be linked to a record at the policy level:
   a. One or more risks
   b. Zero or more forms
   c. Zero or more coverages
2. **Risk** – Each risk represents a risk table in the rating content database. Examples of risks include vehicles on a Commercial Auto policy or locations and buildings on a Businessowners policy. There can also be zero or more child risks associated with a parent risk. The following items can be linked to each record at the risk level:
   a. Zero or more child risks
   b. Zero or more forms
   c. Zero or more coverages
   d. Zero or more schedules (not shown in diagram for clarity)
3. **Form** - Each form represents a form table in the rating content database. There can be zero or more forms associated with a policy record or with a risk. Examples of forms include Audio Equipment and Driver Exclusion on a Commercial Auto policy. The following items can be linked to each record at the form level:
   a. Zero or more coverages
   b. Zero or more schedules
4. **Coverage** - Each coverage represents a coverage table in the rating content database. There can be zero or more coverages associated with a policy record, a risk, or a form. Examples of coverages include specific protection for Collision, Liability, or Personal Injury on a Commercial Auto policy.
5. **Schedule** - Each schedule represents a schedule table in the rating content database. An example of a schedule is a list of names for Named Individuals -

Personal Injury Protection coverage on a Commercial Auto policy. There can be zero or more schedules associated with a "scheduled" risk or form, and there can be zero or more coverages linked to each record at the schedule level (not shown in diagram for clarity purposes).

Note: In addition to ERC, we have another product called "ERC+software". The following types of tables are only used with ERC+software. Please disregard them for ERC purposes.

1. Human
2. Company
3. Payment
4. Note
5. User
6. Address

DS File - Column Detail Tabs

Each table is assigned columns. Some are system generated and some are coverage specific and added based on the line of business requirements from the ISO CLM. The system generated columns provide insurance-specific support for rating. These include items such as Effective Date, Premium, and Unit Number.

Columns can either be inputs or derived values for use within the algorithm. An example for input would be a class code. In Commercial Auto, a class code is represented as a column on a Truck Risk. The column "PrimaryClassCode" is controlled by a domain table TruckPrimaryClassCode which contains the class codes filed for the line of business in a state.

Columns can also store derived values generated by the algorithm. Continuing with the previous example, the class code can be a determining factor in the loss cost utilized in rating. The class code is a key to the lookup of the appropriate factor used in the rating algorithm. For example, TruckPrimaryClassCode is the key to determining the appropriate factor to be pulled from the loss cost table TruckPrimaryClassLiabilityFactor. This value is then used in the algorithm to develop and populate the column LiabilityPremium on the CommercialAutoVehicleLiabilityCoverage table.

The Column Details tab describes the entire structure of the line of business and provides the type, size, and scale of the column. See the example below followed by a description of each column.

Table	Column	Type	Size	Scale	Comment
CommercialPropertyStructure	ProtectionClassToUse	Text	16		
	Office0702ClassCodeRate	Decimal	8	4	
	HighestBusnIncomeBroadBaseRate	Decimal	8	4	
	BldgVacant	Text	3		
	SinkholeLossFactor	Decimal	8	4	
	ExteriorWallsOfClayTileOrConcreteBlocksAndNotFacedWit	Text	80		
	RoofDeckAttachmentAndRoofWallConnectionStatCode	Text	3		
	GreenUpgradeExposureStatCode	Integer			
	Surcharge	Decimal	10	2	
	OrdinanceOrLawCoverageIndicatorNoCoverage	Integer			
	BuildingWithApartmentsIndicator	Text	5		
	BasicGroupIINumericValue	Decimal	8	4	
	LossCostMultiplierStatCode	Text	5		
	SubStdConditionERate	Decimal	8	4	
	TerrorismTerritoryTier	Integer			
	TerrorismBeginYearRate	Decimal	8	4	
	BasicGroupIDeductibleStatCode	Text	2		
	SpecialFinalRate	Decimal	8	4	

There are several types of columns. These are listed and described below.

- **Decimal** – The information is a decimal value (e.g., used with Loss Costs, Rates, or Premiums). With decimal data types, the size and scale values must be specified. The standards are as follows:
  - **Rates and Factors** - 8 size, 4 scale
  - **Premium** - 10 size, 2 scale
- **Integer** - The information is an integer value (e.g., used with limits, number of employees, or payroll). When the integer data type is specified, the size and scale fields are not used.
- **Text** - The information is a text value. When the text data type is specified, the size value must be specified.
- **Unique Identifier** – The information is a unique identifier and used to link tables (e.g., used with Human, Address, or Company).

Rating Content (RC) Tabs

Below is a list and brief description of the tabs within the RC file followed by an illustration of each.

- Ratebook Properties
- Ratebook Hierarchy
- Ratebook Tables or Linked Tables
- Ratebook Columns
- Domain Tables
- Domain Table Keys
- Domain Table Entries
- Rate Tables
- Rate Table Keys
- Rate Table Value Columns
- Rate Table Entries
- Pages
- Page Fields
- Page Related Fields

Ratebook Properties Tab

The Ratebook Properties tab provides an overview of the structured content contained within the file and identifies the Company Name, Locale, Line of Business, State, Effective Date, and Version associated with the content. See the example below followed by a description of the columns within the Ratebook Properties report.

Report Type: Exceptions (Additions And Deletions)								
Company Name	***Locale***	***Line of Business***	***State***	***New Business Effective***	***Renewal Effective***	***Version***	***Published***	***Released***
ISO	English - United States	Commercial Property	Alabama	5/1/2015	5/1/2015	2	2/12/2015 9:29:(	yes

- **Report Type: Exceptions (Additions And Deletions)** – This section indicates that the RC file includes changes – additions and deletions. Every RC file reflects some change to the ISO CLM that has been published and approved in an ISO Circular. Note: Prior to August 2011, all report types were “Full” instead of “Exceptions”. Full report types not only contain unique items per the state manual, but all inherited values, as well:
  - Inherited values are shown in black font

- All unique items are shown in blue font

- **Company Name** – Identifies the source of the information that the rating content is based on. The company name will be "ISO" except with Workers Compensation and a few Commercial Auto packages where the source of information is a rating bureau. For example, rating content for Workers Compensation in Delaware is shown as Delaware Compensation Rating Bureau (DCRB).
- **Locale** - Identifies the language and country of origin for the rating content. This will always be "English - United States".
- **Line of Business** - Identifies the specific line of business that the RC file covers.
- **State** - Identifies the specific state that the RC file covers. Note: In a countrywide file, this column will be blank; if you are working with a multi-state file, "XX" will be displayed.
- **New Business Effective** - Indicates the effective date of the rating content.
- **Renewal Effective Date** - Will always match the "New Business Effective" date.
- **Version** - Identifies the version of the content for the given effective date. Note: If a version is not specified, by default that means you are working with the first version of the rating content; otherwise, you are working with whatever version number is shown in this field.
- **Published and Released** - This information is used with our other product, ERC+software; please disregard this for ERC purposes.

Ratebook Hierarchy Tab

The Ratebook Hierarchy tab outlines the hierarchy of the content in the RC file. Every ERC package includes layers of content including non-LOB base content (information universal to all lines of business), countrywide content, and state content if you are working with a state release. See the example below followed by a description of the columns within the Ratebook Hierarchy report.

Company Name	Locale	Line of Business	State	Title	New Busine Effective
ISO	English - United States	Commercial Property	Alabama	ISO Commercial Property AL	5/1/2015
ISO	English - United States	Commercial Property		ISO Commercial Property CW	5/1/2015
AscendantOne	English - United States			AscendantOne Non-LOB Specific Domains and Constraint	4/15/2004
AscendantOne	English - United States			AscendantOne Non-LOB Specific Domains and Constraint	3/25/2004
AscendantOne	English - United States			AscendantOne Non-LOB Specific Domains and Constraint	4/1/2005
AscendantOne	English - United States			AscendantOne Non-LOB Specific Domains and Constraint	4/2/2003
AscendantOne	English - United States			AscendantOne Non-LOB Specific Domains and Constraint	1/1/2001

- **Company Name, Locale, Line of Business, and State –** Provides the same information provided for these columns in the Ratebook Properties tab. Note: In addition to ISO, "AscendantOne" is identified in the Company Name column. AscendantOne refers to the underlying software product used with ERC.

- **Title** - Illustrates the hierarchy or the layers of rating content. At the top of the hierarchy (read from the bottom up) is the non-LOB base content which contains information universal to all lines of business such as specific domains and constraints. Next is the countrywide layer of content which contains the information in the multistate section of the ISO CLM. The last level is the state rating content which contains the exceptions to the countrywide that are located in the State Exception section of the ISO CLM.
- **New Business Effective, Renewal Effective, Version, and Published** – Provides the same information provided in these columns in the Ratebook Properties tab.
- **New Business Effective** and **Renewal Effective** - Identify the effective date of the rating content in the file which is 05012015 in the above example.
- **Version** - Identifies the version of the rating content. Occasionally a Rating Content package is re-released to address a correction, out-of-sequence endorsement, or additional approval circular. When this occurs, the rating content is assigned a new version. If a version number is not shown in this column, this indicates you are working with version 1; otherwise, you are working with whatever version number is shown in this column.

The other information columns are used with our other product, ERC+software. Please disregard them for ERC purposes.

Ratebook Tables or Linked Tables Tab

The “Ratebook Tables” or “Linked Tables” tab identifies the database tables required in building pages and page fields and in the database tables used with ERC algorithms in the rating process. See the example and a brief description of each column below.

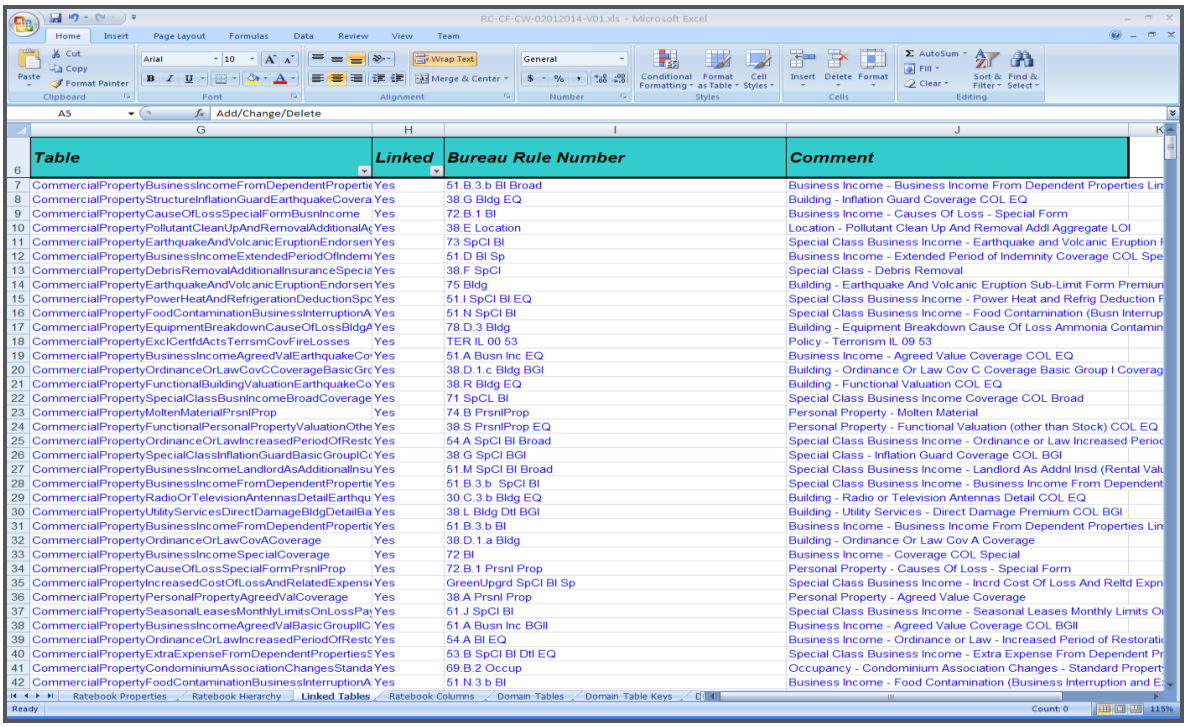


	Table	Linked	Bureau Rule Number	Comment
7	CommercialPropertyBusinessIncomeFromDependentPropertie	Yes	51.B.3.b BI Broad	Business Income - Business Income From Dependent Properties Lin
8	CommercialPropertyStructureInflationGuardEarthquakeCovera	Yes	38.G Bldg EQ	Building - Inflation Guard Coverage COL EQ
9	CommercialPropertyCauseOfLossSpecialFormBusnIncome	Yes	72.B.1 BI	Business Income - Causes Of Loss - Special Form
10	CommercialPropertyPollutantCleanUpAndRemovalAdditionalA	Yes	38.E Location	Location - Pollutant Clean Up And Removal Addl Aggregate LOI
11	CommercialPropertyEarthquakeAndVolcanicEruptionEndorsen	Yes	73 SpCl BI	Special Class Business Income - Earthquake and Volcanic Eruption
12	CommercialPropertyBusinessIncomeExtendedPeriodOfIndemi	Yes	51.D BI Sp	Business Income - Extended Period of Indemnity Coverage COL Spe
13	CommercialPropertyDebrisRemovalAdditionalInsuranceSpecia	Yes	38.F SpCl	Special Class - Debris Removal
14	CommercialPropertyEarthquakeAndVolcanicEruptionEndorsen	Yes	75 Bldg	Building - Earthquake And Volcanic Eruption Sub-Limit Form Premiun
15	CommercialPropertyPowerHeatAndRefrigerationDeductionSpc	Yes	51.I SpCl BI EQ	Special Class Business Income - Power Heat and Refrig Deduction F
16	CommercialPropertyFoodContaminationBusinessInterruptionA	Yes	51.N SpCl BI	Special Class Business Income - Food Contamination (Busn Interrup
17	CommercialPropertyEquipmentBreakdownCauseOfLossBldgA	Yes	78.D.3 Bldg	Building - Equipment Breakdown Cause Of Loss Ammonia Contamin
18	CommercialPropertyExclCertfdActsTerrsmCovFireLosses	Yes	TER IL 00 53	Policy - Terrorism IL 09 53
19	CommercialPropertyBusinessIncomeAgreedValEarthquakeCo	Yes	51.A Busn Inc EQ	Business Income - Agreed Value Coverage COL EQ
20	CommercialPropertyOrdinanceOrLawCovCCoverageBasicGrc	Yes	38.D.1.c Bldg BGI	Building - Ordinance Or Law Cov C Coverage Basic Group I Coverag
21	CommercialPropertyFunctionalBuildingValuationEarthquakeCo	Yes	38.R Bldg EQ	Building - Functional Valuation COL EQ
22	CommercialPropertySpecialClassBusnIncomeBroadCoverage	Yes	71 SpCL BI	Special Class Business Income Coverage COL Broad
23	CommercialPropertyMoltenMaterialPrsnlProp	Yes	74.B PrsnlProp	Personal Property - Molten Material
24	CommercialPropertyFunctionalPersonalPropertyValuationOthe	Yes	38.S PrsnlProp EQ	Personal Property - Functional Valuation (other than Stock) COL EQ
25	CommercialPropertyOrdinanceOrLawIncreasedPeriodOfRestc	Yes	54.A SpCl BI Broad	Special Class Business Income - Ordinance or Law Increased Period
26	CommercialPropertySpecialClassInflationGuardBasicGroupIC	Yes	38.G SpCl BGI	Special Class - Inflation Guard Coverage COL BGI
27	CommercialPropertyBusinessIncomeLandlordAsAdditionalInsu	Yes	51.M SpCl BI Broad	Special Class Business Income - Landlord As Addnl Insd (Rental Valu
28	CommercialPropertyBusinessIncomeFromDependentPropertie	Yes	51.B.3.b SpCl BI	Special Class Business Income - Business Income From Dependent
29	CommercialPropertyRadioOrTelevisionAntennasDetailEarthqu	Yes	30.C.3.b Bldg EQ	Building - Radio or Television Antennas Detail COL EQ
30	CommercialPropertyUtilityServicesDirectDamageBldgDetailBa	Yes	38.L Bldg Dtl BGI	Building - Utility Services - Direct Damage Premium COL BGI
31	CommercialPropertyBusinessIncomeFromDependentPropertie	Yes	51.B.3.b BI	Business Income - Business Income From Dependent Properties Lin
32	CommercialPropertyOrdinanceOrLawCovACoverage	Yes	38.D.1.a Bldg	Building - Ordinance Or Law Cov A Coverage
33	CommercialPropertyBusinessIncomeSpecialCoverage	Yes	72 BI	Business Income - Coverage COL Special
34	CommercialPropertyCauseOfLossSpecialFormPrsnlProp	Yes	72.B.1 Prsnl Prop	Personal Property - Causes Of Loss - Special Form
35	CommercialPropertyIncreasedCostOfLossAndRelatedExpens	Yes	GreenUpgrd SpCl BI Sp	Special Class Business Income - Incrd Cost Of Loss And Reltd Expn
36	CommercialPropertyPersonalPropertyAgreedValCoverage	Yes	38.A Prsnl Prop	Personal Property - Agreed Value Coverage
37	CommercialPropertySeasonalLeasesMonthlyLimitsOnLossPay	Yes	51.J SpCl BI	Special Class Business Income - Seasonal Leases Monthly Limits O
38	CommercialPropertyBusinessIncomeAgreedValBasicGroupIIC	Yes	51.A Busn Inc BGII	Business Income - Agreed Value Coverage COL BGII
39	CommercialPropertyOrdinanceOrLawIncreasedPeriodOfRestc	Yes	54.A BI EQ	Business Income - Ordinance or Law - Increased Period of Restorati
40	CommercialPropertyExtraExpenseFromDependentPropertiesS	Yes	53.B SpCl BI Dtl EQ	Special Class Business Income - Extra Expense From Dependent Pr
41	CommercialPropertyCondominiumAssociationChangesStanda	Yes	69.B.2 Occup	Occupancy - Condominium Association Changes - Standard Propert
42	CommercialPropertyFoodContaminationBusinessInterruptionA	Yes	51.N.3.b BI	Business Income - Food Contamination (Business Interruption and E

- **Table** - Identifies the name of the table.
- **Linked** – Identifies “Yes” or “No” indicating whether the table is utilized within the specific ERC package. The reason why all the tables are marked “Yes” is because behind the scenes there is an underlying database (i.e., the DS file) which contains an inventory of all the tables for a line of business including active and obsolete tables at the countrywide and state levels. The tables marked with a “Yes” as shown in the above example are extracted from this database because these are the tables used with the specific ERC package.
- **Bureau Rule Number** - The rule in the ISO CLM that applies to this table.
- **Comment** – Identifies the name of the table in the ALG Table of Contents. Note: there may exceptions in which case these would be identified in the AG file in the Rule Number Guideline tab.

Ratebook Columns Tab

There is a great deal of information provided within ERC, but not all the data is required to rate a policy. The purpose of the Ratebook Columns tab is to identify the specific information that is required for rating purposes. See the example and a brief description of each column below.

Table	Column	Rating Required	Rating Required Condition	Comment	Status
CommercialProperty	ConditionalExclusionRatingOption	Yes			Added
	CoverageForCertifiedActs	Yes			Added
	CoverageForOtherActs	Yes			Added
	PostTRIAExclusion	Yes			Added
CommercialPropertyEarthquakeAndVolcanicEruptionEndors	EQSubLimitPercent	Yes	EQCovIndicator [.='Yes, Not Included in Blanket']		Added
	SubLimitPct	Yes			Deleted
CommercialPropertyEarthquakeAndVolcanicEruptionEndors	EQSubLimitPercent	Yes	EQCovIndicator [.='Yes, Not Included in Blanket']		Added
	SubLimitPct	Yes			Deleted
CommercialPropertyEarthquakeAndVolcanicEruptionEndors	EQSubLimitPercent	Yes	EQCovIndicator [.='Yes, Not Included in Blanket']		Added
	SubLimitPct	Yes			Deleted

- **Table** - Lists all the tables involved in rating.
- **Column** - Identifies the specific information within a table that needs to be captured to rate a policy.
- **Rating Required** – Identifies “Yes” or “No” whether the field is required in the rating process. This means that values for these items are expected in the ALG to generate a premium. Note: The only time you are going to see a “No”, is when the value is needed for stat assignment.
- **Rating Required Condition** - Identifies conditions that determine the type of policy that would require this field to be populated for rating purposes.
- **Status** – Identifies whether the column has been defined at the countrywide or state level. This information will be utilized in the creation of a common line of business definition while layering in state specific filing information.

Domain Table Tabs

The “Domain Table” tabs, including “Domain Tables”, “Domain Table Keys”, and “Domain Table Entries”, are all related and work together to provide information about the valid options available for a desired page field.

Domain Tables Tab

The “Domain Tables” tab contains a list of all domain tables applicable to the countrywide or state rating content. See the examples below followed by a brief description of each column within the tab.

Table	Title	Type	Data Type	Size	Scale	Xml Type	Bureau Rule Number	Comment	Masked
AddlInsurableInt	Additional Insurable Interests	picklist	Text	128			66.D.1.b	Legal Liability Fo	No
AddlInsurableIntOption	Additional Insurable Interest Option	picklist	Text	16			66.D.1.b		No
BasicGroupIIRatingTerrAL	Basic Group II Rating Territory AL	picklist	Text	80			Territory		No
BasicGroupIRatingTerr	Basic Group I Rating Territory	picklist	Text	80			LOCATION		No
BasicGroupIRatingTerritoryAL	Basic Group I Rating Territory AL	picklist	Text	128			LC 85		No
CityTownState	City Town State	picklist	Text	128			Location		No
CoinsuranceForOptionalFormsAL	Coinsurance for Optional Forms - Alabama	picklist	Text	16			A4.B Coinsurance		No
ConformsToCondoRule	Conforms To Condominiu Rules - Special Class	picklist	Text	16			69.B.2; 14.E SpCl		No
ConformsToCondoRuleBldg	Conforms To Condominium Rules - Building	picklist	Text	16			69.B.2; 14.E Bldg		No
ConformsToCondoRuleOccupClass	Conforms To Condominiu Rules - Occup Class	picklist	Text	16			69.B.2; 14.E Occup		No
CountyState	County State	picklist	Text	128			Location		No
EQDeductibleBuilding	Earthquake Deductible	picklist	Text	16			73.D.2.d	Building (Structu	No
EQRatingTerrAL	Earthquake Rating Territory AL	picklist	Text	80			Territory		No

- **Table** – Identifies the names of all the domain tables applicable to the rating content.
- **Title** – Identifies the name of the table but with spacing between words.
- **Type –** Provides the types of values available for a page field. There are two types of values:
  1. **Picklist** – The domain table value is defined from a distinct list of choices and has an exact match. See the example below.

CauseOfLoss	Sequence	Display Value	DataValue
	1	Basic	Basic
	2	Broad	Broad
	3	Special	Special

  2. **Range** – The domain table value is between a minimum and maximum value and can identify a set of ranges for an entry. See the example below.

UtilityServicesTimeElementLimit	Sequence	MinimumValue	MaximumValue
	1	1	999999999

- **Data Type** – Identifies the type of data allowed in populating a page field. There are four types of data:
  1. Date
  2. Decimal
  3. Integer
  4. Text
- **Size** – Defines the length of a field or constraint. Note: Size is applicable only when the data type is decimal or text.
- **Scale** – This constraint is only applicable to the decimal data type.
- **Masked** – Indicates whether the domain table is masked. Note: A masked table changes the structure of an inherited table with the same name. Masked tables are no longer used and are in the process of being removed.
- **Bureau Rule Number** – Identifies the rule in the ISO CLM that is being addressed.

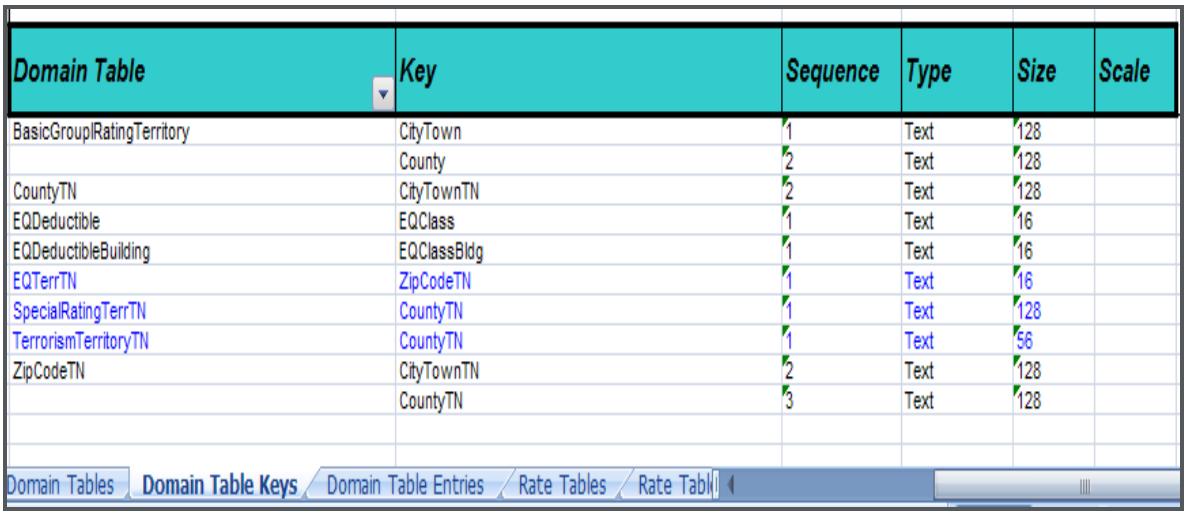
The other information columns are used with our other product, ERC*+software*, and should be disregarded for ERC purposes.

Domain Table Keys Tab

The “Domain Table Keys” tab provides the criteria for selections within a given table. If a domain table does not have an associated key, all the values listed in the domain

table will be available. If a domain table has one or more keys, the values that are available are only those that match the data in the key(s).

The Domain Table Keys tab will only contain the domain tables that have at least one key. See the example below including a brief description of each column.



Domain Table	Key	Sequence	Type	Size	Scale
BasicGroupIRatingTerritory	CityTown	1	Text	128	
	County	2	Text	128	
CountyTN	CityTownTN	2	Text	128	
EQDeductible	EQClass	1	Text	16	
EQDeductibleBuilding	EQClassBldg	1	Text	16	
EQTerrTN	ZipCodeTN	1	Text	16	
SpecialRatingTerrTN	CountyTN	1	Text	128	
TerrorismTerritoryTN	CountyTN	1	Text	56	
ZipCodeTN	CityTownTN	2	Text	128	
	CountyTN	3	Text	128	

Domain Tables | Domain Table Keys | Domain Table Entries | Rate Tables | Rate Tabl

- **Domain Table** - Lists the names of all the domain tables that have keys.
- **Key** - Names the specific key or constraint within the domain table.
- **Sequence** - Represents the order in which the keys will be used in the domain table.
- **Type** - Provides the types of keys -- date, decimal, integer, and text.
- **Size** - Defines the length of the key which is required only when the data type is decimal or text.
- **Scale** - Applies only to the decimal data type.

The other information columns are used with our ERC*+software* product, and should be disregarded for ERC purposes.

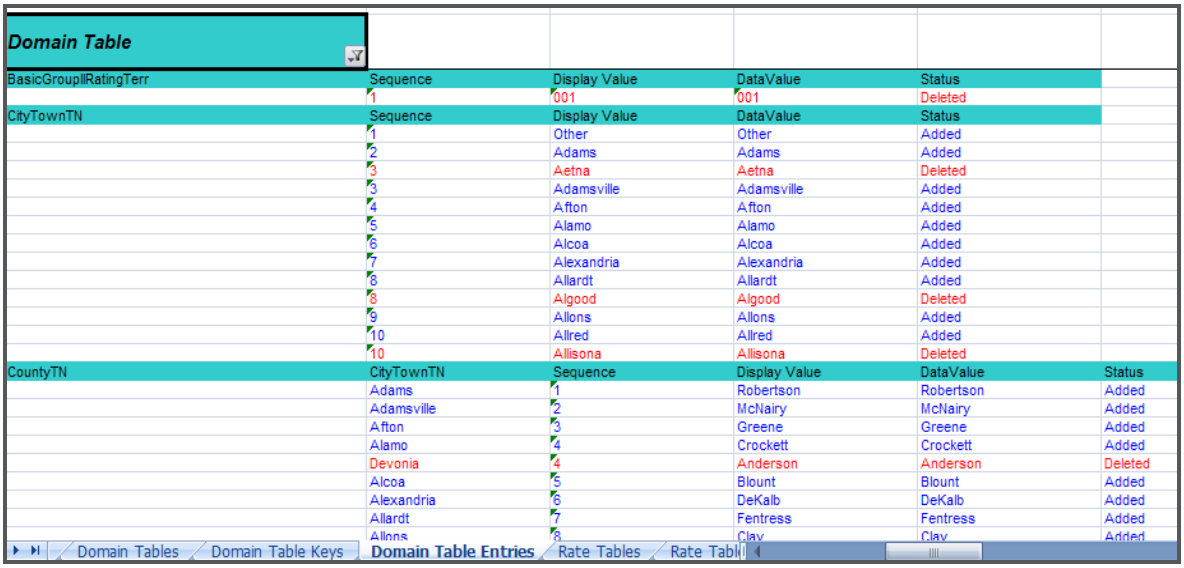
In the above example, the ZipCodeTN domain table has two keys -- CityTown and County. The data for City and County needs to be provided to determine the values that are available for the ZipCode result. Based on this example and the table below, we can determine that when the city is Angola and the county is Steuben, then the ZipCode values available to select are only: 46703 and Other.

ZipCodeIN	CityTownIN	CountyIN	Sequence	Display Value	DataValue
	Anderson	Madison	28	46011	46011
	Anderson	Madison	29	46012	46012
	Anderson	Madison	30	46013	46013
	Anderson	Madison	31	46014	46014
	Anderson	Madison	32	46015	46015
	Anderson	Madison	33	46016	46016
	Anderson	Madison	34	46017	46017
	Anderson	Madison	35	46018	46018
	Anderson	Madison	36	Other	Other
	Anderson	Other	37	Other	Other
	Andrews	Huntington	38	46702	46702
	Andrews	Huntington	39	Other	Other
	Andrews	Other	40	Other	Other
	Angola	Steuben	41	46703	46703
	Angola	Steuben	42	Other	Other
	Angola	Other	43	Other	Other

Note: The data value “Other” means the user must key in the zip code. This option is used with our ERC+software product and should be disregarded for ERC purposes.

Domain Table Entries Tab

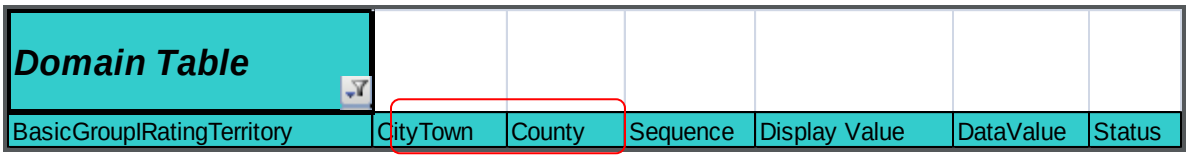
The “Domain Table Entries” tab lists all the data entry selections associated with the applicable domain table listed in the “Domain Tables” tab. See the example below followed by a brief description of each column.



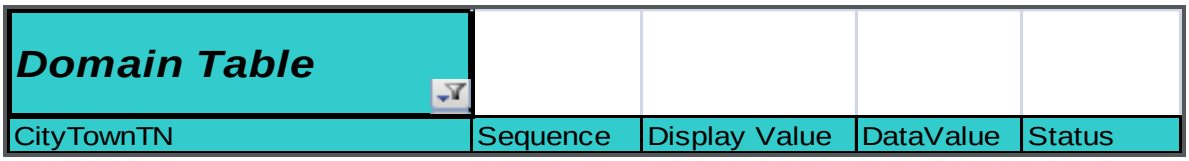
Domain Table					
BasicGroupIIRatingTerr	Sequence	Display Value	DataValue	Status	
	1	001	001	Deleted	
CityTownTN	Sequence	Display Value	DataValue	Status	
	1	Other	Other	Added	
	2	Adams	Adams	Added	
	3	Aetna	Aetna	Deleted	
	3	Adamsville	Adamsville	Added	
	4	Afton	Afton	Added	
	5	Alamo	Alamo	Added	
	6	Alcoa	Alcoa	Added	
	7	Alexandria	Alexandria	Added	
	8	Allardt	Allardt	Added	
	8	Algood	Algood	Deleted	
	9	Allons	Allons	Added	
	10	Allred	Allred	Added	
	10	Allisona	Allisona	Deleted	
CountyTN	CityTownTN	Sequence	Display Value	DataValue	Status
	Adams	1	Robertson	Robertson	Added
	Adamsville	2	McNairy	McNairy	Added
	Afton	3	Greene	Greene	Added
	Alamo	4	Crockett	Crockett	Added
	Devonia	4	Anderson	Anderson	Deleted
	Alcoa	5	Blount	Blount	Added
	Alexandria	6	DeKalb	DeKalb	Added
	Allardt	7	Fentress	Fentress	Added
	Allons	8	Clay	Clay	Added

Domain Tables | Domain Table Keys | Domain Table Entries | Rate Tables | Rate Tabl

- **Key(s)** - If a domain table has one or more keys, then the keys columns will be displayed before any other element. If the domain table doesn't have any keys, the next element will be displayed. See the example below where Domain Table BasicGroupIRatingTerritory has two keys, CityTown and County. These two keys will be displayed in the order of their sequence from left to right.



Domain Table CityTownTN doesn't have any keys. See the example below.



- **Sequence** – Represents the order of the pick list for display purposes.
- **DisplayValue** – This is used for internal purposes; please disregard.
- **DataValue** – This is for a picklist domain table and represents the collected value. The type of data in this column is the data type selected for the Domain Table.
- **MinimumValue** – This is used for a range domain table and contains the minimum value of the defined range. The type of the data in this column is the data type selected for the Domain Table.
- **MaximumValue** – This is for a range domain table and contains the maximum value of the defined range. The type of the data in this column is the data type selected for the Domain Table
- **Status** – Indicates whether an entry has been added, changed, or deleted from the table. Once an item is either added or changed, it remains permanent with that status until it is either deleted or re-inherited, i.e., from the countrywide. Once an item is deleted, it remains permanent with that status until it is re-inherited from a base or parent report.

Domain Tables – Relationship to ISO CLM

Within the rating content there are domain tables that are based on specifications within a rule. For example, the rule table 85.M.1.b. Protection Class Multipliers in Commercial Property Illinois has the following options for Protection Class: 1, 2, 3, 4, 5, 6, 7, 8, 8B, 9 and 10. See the example below.

Protection Class	Frame, Joisted Masonry, Non-Combustible	Masonry Non-Combustible, Modified Fire Resistive, Fire Resistive
1	.88	.90
2	.92	.94
3	.96	.97
4	.98	.98
5	1.00	1.00
6	1.06	1.05
7	1.18	1.14
8	1.30	1.24
8B	1.42	1.34
9	1.42	1.34
10	1.72	1.58

In the Illinois rating content, we have a domain table called “ProtectionClass” which would contain the values listed on the rule table. See the example below.

ProtectionClass	Sequence	Display Value	DataValue
	1	01	01
	2	02	02
	3	03	03
	4	04	04
	5	05	05
	6	06	06
	7	07	07
	8	08	08
	9	8B	8B
	10	09	09
	11	10	10

There are other generic domain tables that can be used for different fields. For example, if a field has the value of “Yes” or “No”, this field would use the “YesNo” generic domain table containing the Yes and the No values. See the example below.

YesNo	Sequence	Display Value	DataValue
	1	No	No
	2	Yes	Yes

Domain Tables – Relationship to Page Fields

Domain tables are created to be used with page fields. We can categorize the relationship between domain tables and page fields as follows:

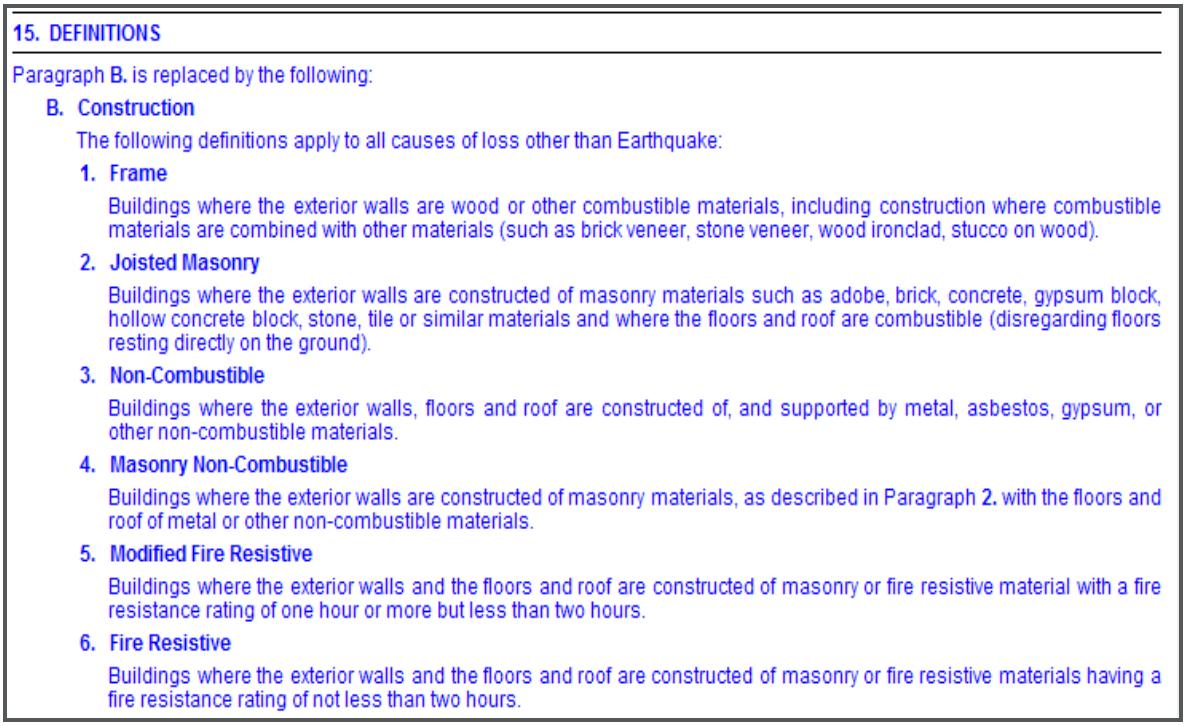
- **One to One** - One domain table is assigned to a page field.

- **One to Many** - One domain table is assigned to various page fields.

However, each page field can only be linked to one domain table. Most fields in rating content are associated with a domain table.

Some page fields utilize range domain tables for restricting input of a decimal or an integer. For example, a Limit field that requires the input to be more than $100000 would use a range domain to restrict the value of the field to at least $100000.

A good example of how keys in a domain table work based on another page field is in obtaining the correct Construction Code after selecting a Construction Type. In Commercial Property, the Washington rule exception 15.B has a list of construction definitions, as follows:



15. DEFINITIONS

Paragraph **B.** is replaced by the following:

B. Construction

The following definitions apply to all causes of loss other than Earthquake:

1. **Frame**

   Buildings where the exterior walls are wood or other combustible materials, including construction where combustible materials are combined with other materials (such as brick veneer, stone veneer, wood ironclad, stucco on wood).

2. **Joisted Masonry**

   Buildings where the exterior walls are constructed of masonry materials such as adobe, brick, concrete, gypsum block, hollow concrete block, stone, tile or similar materials and where the floors and roof are combustible (disregarding floors resting directly on the ground).

3. **Non-Combustible**

   Buildings where the exterior walls, floors and roof are constructed of, and supported by metal, asbestos, gypsum, or other non-combustible materials.

4. **Masonry Non-Combustible**

   Buildings where the exterior walls are constructed of masonry materials, as described in Paragraph **2.** with the floors and roof of metal or other non-combustible materials.

5. **Modified Fire Resistive**

   Buildings where the exterior walls and the floors and roof are constructed of masonry or fire resistive material with a fire resistance rating of one hour or more but less than two hours.

6. **Fire Resistive**

   Buildings where the exterior walls and the floors and roof are constructed of masonry or fire resistive materials having a fire resistance rating of not less than two hours.

The ConstructionType page field uses a picklist domain table which contains all the types of constructions listed in the ISO CLM. See the example below.

ConstructionTypeBldg	Sequence	Display Value	DataValue
	1	Fire Resistive	Fire Resistive
	2	Frame	Frame
	3	Joisted Masonry	Joisted Masonry
	4	Masonry Non-Combustible	Masonry Non-Combustible
	5	Modified Fire Resistive	Modified Fire Resistive
	6	Non-Combustible	Non-Combustible

The value then selected for ConstructionType is passed as a key on the ConstructionCode domain table. Depending on the construction type, the domain table will return the code listed for that type. Then the code from the DataValue column would be assigned to the ConstructionCode page field. See the example below.

ConstructionCode	ConstructionTypeToUse	Sequence	Display Value	DataValue
	Frame	1	1	1
	Joisted Masonry	2	2	2
	Non-Combustible	3	3	3
	Masonry Non-Combustible	4	4	4
	Modified Fire Resistive	5	5	5
	Fire Resistive	6	6	6

Domain Tables – Relationship to Rate Tables

Domain table data values are very important in relation to rate tables. Rate table criteria selections are based upon domain table entries. Most of the domain table entries are used as keys within rate tables. For example, in Commercial Property under the Occupancy Class risk level the ClassCode page field uses a domain table called ClassCodeOccupClass and the OccupCategory page field uses a Domain table called OccupancyCategory. See the example below.

Domain Table

ClassCodeOccupClass	RatingType	Sequence	Display Value	DataVal	Status
	Class	1	0074	0074	Added
	Class	2	0075	0075	Added
	Class	3	0076	0076	Added
	Specific	100	0074	0074	Added
	Specific	101	0075	0075	Added
	Specific	102	0076	0076	Added
	Tentative	236	0074	0074	Added
	Tentative	237	0075	0075	Added
	Tentative	238	0076	0076	Added
OccupancyCategory	Sequence	Display Value	DataValue	Status	
	1	Residential Apartments and Condominiums	Residential Apartments and Condominiums	Added	
	2	Offices	Offices	Added	
	3	Mercantile	Mercantile	Added	
	4	Motels and Hotels	Motels and Hotels	Added	
	5	Institutional	Institutional	Added	
	6	Industrial and Processing	Industrial and Processing	Added	
	7	Service	Service	Added	
	8	Contractors	Contractors	Added	

A value from these domain tables will be assigned to their respective page fields. The values from these page fields will then be used to determine the value of OccupCategoryRiskSeverity page field through a rate table lookup.

If the value selected for the ClassCode page field is “0074” and the value selected for the OccupCategory page field is “Offices”, then the value that the rate table lookup will return for OccupCategoryRiskSeverity page field will be “Not Applicable”. See the example below.

Rate Table				
OccupCategoryRiskSeverity	OccupCategory	ClassCode	Severity	Status
	Contractors	0074	Not Applicable	Added
	Contractors	0075	Not Applicable	Added
	Contractors	0076	Not Applicable	Added
	Industrial and Processing	0074	Low	Added
	Industrial and Processing	0075	Low	Added
	Industrial and Processing	0076	Low	Added
	Institutional	0074	Low	Added
	Institutional	0075	Low	Added
	Institutional	0076	Low	Added
	Mercantile	0074	Medium	Added
	Mercantile	0075	Medium	Added
	Mercantile	0076	Medium	Added
	Motels and Hotels	0074	Not Applicable	Added
	Motels and Hotels	0075	Not Applicable	Added
	Motels and Hotels	0076	Not Applicable	Added
	Offices	0074	Not Applicable	Added
	Offices	0075	Not Applicable	Added
	Offices	0076	Not Applicable	Added
	Residential Apartments and Condominiums	0074	Not Applicable	Added
	Residential Apartments and Condominiums	0075	Not Applicable	Added
	Residential Apartments and Condominiums	0076	Not Applicable	Added
	Service	0074	Low	Added
	Service	0075	Low	Added
	Service	0076	Low	Added

Rate Tables Tabs

The Rate Tables reports -- Rate Tables, Rate Table Keys, Rate Table Value Column, and Rate Table Entries -- contain all the information used to compute a rate, loss cost, factor or a stat code.

Rate Tables Tab

The Rate Tables tab contains two or more columns, one or more key columns, and a value column. See the example below.

Rate Table			
BroadFormBaseRate	ConstructionTypeToUse	CovType	Rate
	Frame	Bldg	0.015
	Joisted Masonry	Bldg	0.015
	Non-Combustible	Bldg	0.015
	Masonry Non-Combustible	Bldg	0.015
	Modified Fire Resistive	Bldg	0.01
	Fire Resistive	Bldg	0.01
	Frame	PersProp	0.021
	Joisted Masonry	PersProp	0.021
	Non-Combustible	PersProp	0.021
	Masonry Non-Combustible	PersProp	0.021
	Modified Fire Resistive	PersProp	0.011
	Fire Resistive	PersProp	0.011

In this example, the key column is named ConstructionTypeToUse and the value column is named Rate. To compute the rate, a particular construction is looked up in the ConstructionTypeToUse column and then the matching rate value is pulled from the BroadFormBaseRate that is to be computed.

Rate Table Keys

Rate Table Keys determine the appropriate match for a loss cost lookup called from an algorithm. Keys must begin with a letter and may include letters, numbers, or currency symbols. A rate table is required to have at least one key. Generally, keys are matched values from a domain table.

As described in the Domain Table tab section, these entries are very important in relation to our rate tables. Our rate table lookup always passes a parameter; the value in most cases is obtained from a field that uses a domain table. The lookup method of a rate table uses the data value that is passed by the domain table through the page field to return the correct value from the table.

The Rate Table Key must have a valid data type. The following are valid data types:

- **Date** - The key field has a date value. When the date data type is specified, the size and scale fields are not used.
- **Decimal** - The key field has a decimal value. When the decimal data type is specified, the size value must be specified, and the scale value may be specifically optionally.
- **Integer** - The key field has an integer value. When the integer data type is specified, the size and scale fields are not used. The integer data type will store a value between -2,147,483,648 and 2,147,483,647.
- **Long** - The key field has a long value. When the long data type is specified, the size and scale fields are not used. The long data type will store a value between -9,223,372,036,854,775,808 and 9,223,372,036,854,775,807.
- **Text** - The key field has a text value. When the text data type is specified, the size value must be specified.

Note: When a domain table is specified as a key in a rate table, the domain table name is the same as the name of the rate table key. In general, when a rate table key is a domain table, the data type of the key and the domain table must be the same. However, when creating a rate table key that is a domain table with a range type, the

rate table key must have a data type of text, regardless of the data type of the domain table.

Rate Table	Key	Sequence	Type	Size	Scale	Lookup Type
BasicGroupIIRate	BasicGroupIIRatingTerr	1	Text	80		ExactMatch
	BasicGroupIISymbol	2	Text	16		ExactMatch
	CovType	3	Text	16		ExactMatch

Rate Table Value Columns Tab

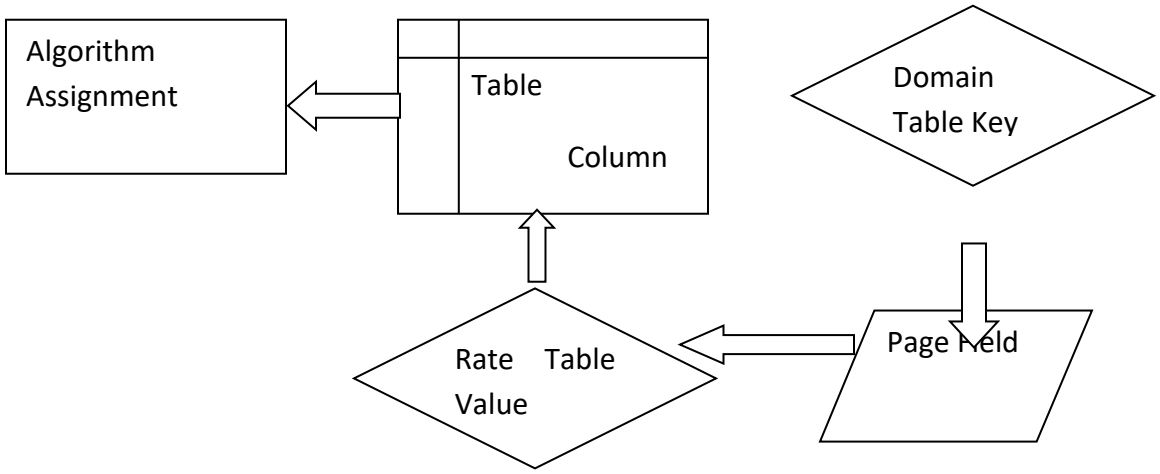
Rate Table Entry Column values also require a data type. The column value field has a name such as factor. The value name must begin with a letter and may include letters, numbers, or currency symbols. The following are the possible data types for the value field:

- **Date** - The field contains a date value. When the date data type is specified, the size and scale fields are not used.
- **Decimal** - The field contains a decimal value. When the decimal data type is specified, the size value must be specified, and the scale value may be specified on an optional basis. If optional, then the scale of the data value may be set if the type is set to decimal.
- **Integer** - The field contains an integer value. When the integer data type is specified, the size and scale fields are not used. The integer data type stores a value between -2,147,483,648 and 2,147,483,647.
- **Long** - The field contains a long value. When the long data type is specified, the size and scale fields are not used. The long data type stores a value between -9,223,372,036,854,775,808 and 9,223,372,036,854,775,807.
- **Text** - The field contains a text value. When the text data type is specified, the size value must also be specified.
- **Size** - The length of the data value is specified in the size field if the type is set to decimal or text.

Rate Table	Column	Type	Size	Scale
BaseRateAdjustmentFactor	Factor	Decimal	8	4
BasicGroupIIRate	Rate	Decimal	8	4
BasicGroupIRate	Rate	Decimal	8	4
BCEGEQFactor	Factor	Decimal	8	4
BCEGFactor	Factor	Decimal	8	4
BldgHeightFactor	Factor	Decimal	8	4
BurglaryAlarmCreditFactor	Factor	Decimal	8	4
BurgRobbery1FactorMO	Factor	Decimal	8	4
BurgRobbery2FactorMO	Factor	Decimal	8	4
EarthquakeDeductibleTier	Value	Integer		

The result of a rate table lookup value is placed in a column determined by the structure of the line of business. This column is then called in a specific order of operation within a coverage algorithm.

Process Example:



Following is an example of the process used with rate tables using Rule 41.D, Collapse During Construction, as an illustration. We can see that the Collapse options available to select are Options 1, 2, and 3. The rate is looked up from the BldrsRiskCollapseDuringConstrctnRate rate table. Following the algorithm, you will see that the rate determined will then be subtracted by 1 before being multiplied by the Broad Rate on the Commercial Property Structure table.

Algorithm	Rule Reference
Building Total Premium	Total Building Premium
Building Common Rating	Building Common
Building - Additional Covered Property	Bldg_AddlCoveredProp
Building Builders Risk Collapse During Construction	Rule 41.D
Building Builders Risk Renovations	Rule 41.A

Type	Operator	Value
		Rate
Assignment		Rate
	=	Rate Table: Bldrs Risk Collapse During Constrctn Rate(Using Collapse Option

Rate Table

BldrsRiskCollapseDurin	BldrsRiskCollapseDuringCons	Rate	Status
	Option1	5	Added
	Option2	5	Added
	Option3	0.002	Added

	Rule#	Name
Primary Rule	Rule 41.D	Commercial Property Builders Risk Collapse During Construction.Premium
DBTablesUsed		Commercial Property Builders Risk Collapse During Construction
Total Premium		Commercial Property Builders Risk Collapse During Construction
Type	Operator	Value
		Premium
Assignment		Premium
	=	Calc Premium
		Calc Premium
Assignment		Calc Premium
	=	
	(	
	(	
	(	
	(	Commercial Property Structure\Broad Rate
	*	
	(	Rate
	-	1
	)	
	) Round Up Thousandth	
	*	Commercial Property Structure\Limit To Use
	/	100.0
	) Round Up Dollar	
	*	Commercial Property \ Package Mod Factor
	) Round Up Dollar	
	*	Commercial Property \ IRPM Factor
	) Round Up Dollar	

Rate Table Entries

Rate table entries are derived by look-up methods. This involves placing the appropriate matched value from the rate table entries to the designated column on a line of business table. The following are valid lookup types:

- **Exact Match -** The key field has a lookup type of Exact Match. When Exact Match is specified, a particular value is looked up in the key column and the matching value is added to the rate that is to be computed.
- **Less Than** - The key field has a lookup type of Less Than. When Less Than is specified, a particular value is looked up in the key column and matched with the row containing the highest value that is less than the value being looked up.
- **Less than or Equal** - The key field has a lookup type of Less than or Equal. When Less than or Equal is specified, a particular value is looked up in the key column and then matched with the row containing the highest value that is less than or equal to the value being looked up.
- **Greater Than** - The key field has a lookup type of Greater Than. When Greater Than is specified, a particular value is looked up in the key column and matched with the row containing the lowest value that is greater than or the value being looked up.
- **Greater Than or Equal** - The key field has a lookup type of Greater than or Equal. When Greater than or Equal is specified, a particular value is looked up in the key column and matched with the row containing the lowest value that is greater than or equal to the value being looked up.
- **Interpolate** - The key field has a lookup type of Interpolate. When Interpolate is specified, a particular value is looked up in the key column and then the matching value is added to the rate that is to be computed. In the event that the value falls between two entries in the rate table the return value is determined by interpolating between the two rates in the table.

Key	Factor
0	1.84
1	5.51
4	11.40
9	29.41

Lookup Type	Strategy	If key = 3 (3 is the value being passed)
Exact Match	The rate table key field value must exactly match the value being passed into the rate table from the rate request.	Error - Lookup will fail.
Less Than	The system checks if the value being passed is less than the lowest value in the table and then works its way up to compare against the next highest value in the table.	is value of 3 less than 0 - No is value of 3 less than 1 - No is value of 3 less than 4 - Yes, return 11.4
Less Than or Equal	The system checks if the value being passed is less than or equal to the lowest value in the table and then works its way up to check against the next highest value in the table.	is value of 3 less than or equal to 0 - No is value of 3 less than or equal to 1 - No is value of 3 less than or equal to 4 - Yes, return 11.4
Greater Than	The system checks if the value being passed is greater than the highest value in the table and then works its way down to check again the next lowest value in the table.	is value of 3 greater than 9 - No is value of 3 greater than 4 - No is value of 3 greater than 1 - Yes, returns 5.51
Greater Than or Equal	The system checks if the value being passed is greater than or equal to the highest value in the table and then works its way down to check again the next lowest value in the table.	is value of 3 greater than or equal to 9 - No is value of 3 greater than or equal to 4 - No is value of 3 greater than or equal to 1 - Yes, returns 5.51
Interpolate	The system performs linear interpolation.	((3 - 1)* (11.40 - 5.51)/(4 - 1)) + 5.51 = (2 x 5.89 / 3) + 5.51 = 9.44 (rounded to 2 decimals in this example)

Rate Table	Key	Sequence	Type	Size	Scale	Lookup Type
BasicGroupIILOIFactorBldg	Limit	1	Integer			Interpolate
BasicGroupIILOIFactorPersProp	Limit	1	Integer			Interpolate
BasicGroupIIRate	BasicGroupIIRatingTerr	1	Text	80		ExactMatch
	BasicGroupIISymbol	2	Text	16		ExactMatch
	CovType	3	Text	16		ExactMatch

Rate Table		
BasicGroupIILOIFactorBldg	Limit	Factor
	0	1.513
	50000	1.513
	75000	1.363
	100000	1.266
	125000	1.195
	150000	1.14
	175000	1.096
	200000	1.059
	225000	1.027
	250000	1
	275000	0.976
	300000	0.954
	325000	0.935
	350000	0.917
	375000	0.901
	400000	0.886
	425000	0.872
	450000	0.86
	475000	0.848
	500000	0.837

Page Tabs

The Page tabs – Page, Page Fields, and Page Related Fields -- represent what is displayed on a screen when a page is loaded into a rating system. For example, what fields are on a page, and what are the size and type of those fields.

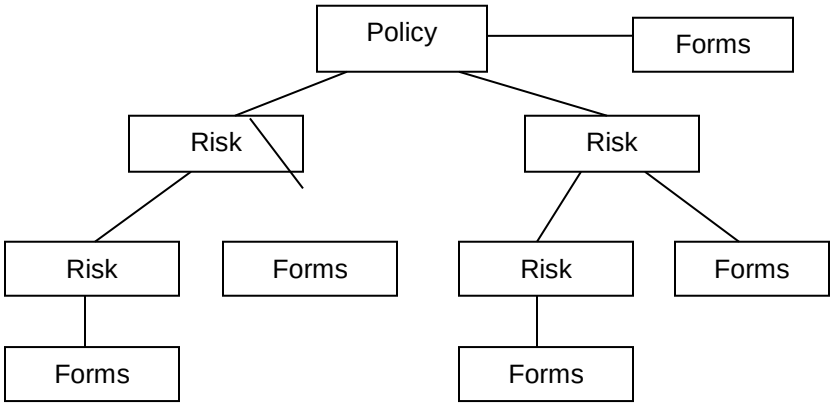
Page Tab

A page has three main types and can be associated with a policy, risk, or a form. Each page is associated with a table; this table links the rating content to the algorithms. Every page can only be associated with a single table. As a result, this relationship between pages and tables contains the hierarchy structure for the state rating content.

For a given line of business, there is one policy page. As a child of this policy page, there can be either risk or form pages. To identify what tables are risks versus forms, apply a filter so only the Policy or Policy Detail columns are displayed. Once this filter is applied, go to the “Type” column and select “Summary”; the result will provide a list of

all child risks of the policy. If the parent was changed as a result of the first filter, the result is the second tier of risk pages within the state content. This process can continue until all risk pages are determined. The final product is a wireframe guide for the state. This same process may be executed for forms; the only difference is under the filter criteria would be “Form“ not “Summary”. The majority of pages within the rating content are forms.

Below is a sample hierarchy for pages.



Below is an example of the Page tab followed by a description of the information it provides:

Name	Parent	Table	Type	Sequence	SubSequence	Form Type	Number
Policy Detail		CommercialProperty	Detail	1000	1		
Premium Adjustment Form	Policy Detail	CommercialPropertyPremiumA	Form	1020	1	Optional	
Policy Changes	Policy Detail	CommercialPropertyPolicyChar	Form	1100	1	Optional	IL 12 01 11 85
Commercial Property Coverage Part	Policy Detail	CommercialPropertyCoverageF	Form	1150	1	Optional	CP 12 01 11 85
Blanket Insurance	Policy Detail	CommercialPropertyBlanketRal	Form	1200	1	Conditional	
Blanket Insurance Summary	Blanket Insurance	CommercialPropertyBlanketRal	Summary	1201	1		
Blanket Insurance Detail	Blanket Insurance Summary	CommercialPropertyBlanketRal	Detail	1202	1		
Calculation Of Premium	Policy Detail	CommercialPropertyCalculation	Form	1225	1	Mandatory	IL 00 03 09 08
Higher Limits	Policy Detail	CommercialPropertyHigherLimi	Form	1234	1	Conditional	CP 04 08 10 12
Cap On Losses From Certified Acts C	Policy Detail	CommercialPropertyCapOnLos	Form	1250	1	Conditional	IL 09 52 01 15
Changes - Fungus, Wet Rot, Dry Rot	Policy Detail	CommercialPropertyChangesF	Form	1251	1	Conditional	CP 04 31 04 02
Exclusion Of Loss Due To Virus Or Ba	Policy Detail	CommercialPropertyExclusionC	Form	1260	1	Mandatory	CP 01 40 07 06
Changes	Policy Detail	CommercialPropertyChanges	Form	1275	1	Mandatory	
Changes - Cancellation And Nonrene	Policy Detail	CommercialPropertyChangesC	Form	1280	1	Mandatory	
Common Policy Conditions	Policy Detail	CommercialPropertyCommonPo	Form	1300	1	Conditional	IL 00 17 11 98
Commercial Property Conditions	Policy Detail	CommercialPropertyCommercia	Form	1400	1	Conditional	CP 00 90 07 88
Disclosure Pursuant To Terrorism Ris	Policy Detail	CommercialPropertyZzDisclosu	Form	1750	1	Conditional	IL 09 85 01 15
Earthquake And Volcanic Eruption Su	Policy Detail	CommercialPropertyEQSubLim	Form	1800	1	Conditional	
Earthquake And Volcanic Eruption Su	Earthquake And Volcanic Eruption Su	CommercialPropertyEQSubLim	Summary	1801	1		

Min Occurs	Max Occurs	Hide Premium	Condition	Variable	Scheduled
0	1	No		N	N
0	99	No	/AscendantOne/Policy	N	N
0	1	Yes		N	N
0	1	Yes		N	N
0	0	Yes		N	N
1	99	No		N	N
0	1	No		N	N
1	1	Yes		N	N
0	0	Yes		N	N
0	0	Yes		N	N
0	0	No		N	N
1	1	Yes		N	N
1	1	Yes		N	N
1	1	Yes		N	N
0	0	Yes		N	N

- **Name -** Represents the name of the page. If a page is a form page, the name is the name of the form.
- **Parent -** Lists the name of the parent page. The parent of a policy page is blank.
- **Table -** This is the name of the table that is associated with the page.
- **Type -** There are three types of pages:
  1. **Form** - The page is a form page, representing a form in the ISO Manual.
  2. **Detail**:
     a. **Policy Detail Page** - Each line of business has one policy page. Policy page is the type detail and is always the first page in sequence.
     b. **Risk Level Detail Page** - Each risk level can have a risk page. Risk page is the type detail.
     c. **Form Detail Page** – The form detail page represents schedule of a corresponding form. It is used to capture data for that form.
  3. **Summary** - Summary pages are used to summarize the key fields of a detail page. Summary pages can be ignored.
- **Form Type -** If a page is a form page, there are three types of form pages:
  1. **Mandatory Forms** – Must be applied to a policy.
  2. **Conditional Forms** – Must be applied to a policy if a condition is met. (The condition can be found in the Algorithms under Form Attachment Logic.)
  3. **Optional Forms** – Overall, optional forms do not have a condition for an attachment, therefore the condition column is blank. However, if an optional form has a condition, then the condition is being used to determine whether or not the form can be used in the policy (eligibility).

  Note: mandatory and conditional forms are found in the ISO CLM as mandatory. Optional forms are found in the ISO CLM as optional.
- **Number -** If the page is a form page, then the form number will be specified. For example, for form CP 00 90 07 88, “CP 00 90” is the form number and “07 88” is the edition date. If the page is a form page and no form number is provided, then this means the page was created for data collection purposes.

- **Min Occurs/Max Occurs** - Indicates the minimum and maximum number of iteration of form availability on a policy:
  - **Mandatory Forms - Min** Occurs = 1, Max Occurs = 1
  - **Conditional Forms** - Min Occurs = 0, Max Occurs = 0 Note: conditional forms depend on certain scenarios or conditions in order to apply to a policy. When a policy is created, the conditional form attachment logic is based on certain user selections. Therefore, the default for min/max is “0/0” because a form can’t be added by a user unless it is determined whether or not a condition is met.
  - **Optional Forms** - Min Occurs = 0, Max Occurs > 99
- **Hide Premium -** This Yes/No indicator identifies whether the form page is premium bearing. Yes means it is not premium bearing; no means it is premium bearing.
- **Condition -** This is the condition for "Optional" forms to be available for selection.
- **Variable –** This identifies forms that have variable versus static data.
- **Scheduled –** Identifies schedule forms.

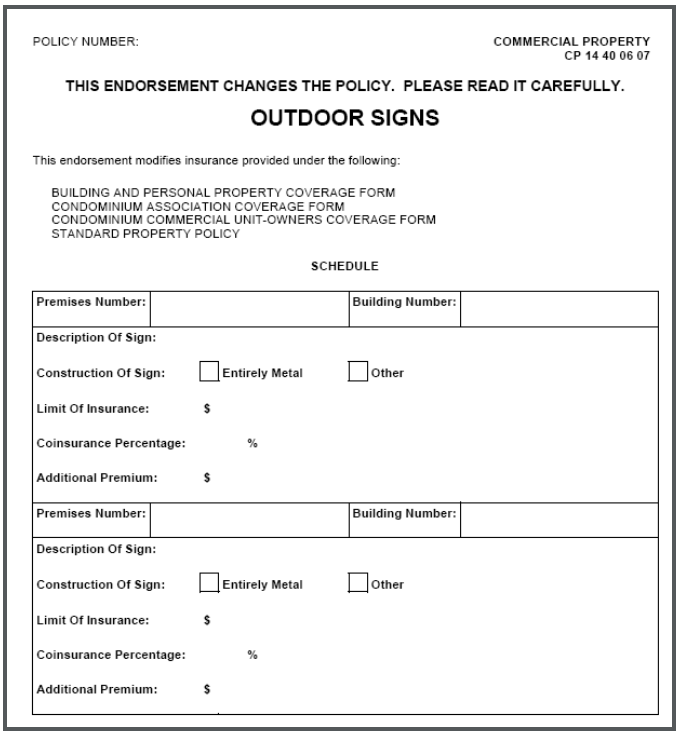
The following list of information is only used with the ERC+software product. As of October 2018, this information is being removed from ERC. If you are working with ERC releases prior to this time, please disregard this information.

1. Sequence
2. Sub Sequence
3. Comment
4. Before Script
5. After Script
6. Template File
7. Tab Condition
8. Tab Resource Name
9. BtnResAdd
10. BtnCndAdd
11. BtnLblAdd
12. BtnResModify
13. BtnCndModify
14. BtnLblModify
15. BtnResDelete
16. BtnCndDelete
17. BtnLblDelete
18. BtnResViewDetail
19. BtnCndViewDetail
20. BtnLblViewDetail
21. BtnResCopy

22. BtnCndCopy
23. BtnLblCopy
24. BtnResRenumber
25. BtnCndRenumber
26. BtnLblRenumber
27. BtnResView
28. BtnCndView
29. BtnLblView

Illustration of ISO Form with Detail Pages

If a form needs to capture additional information for multiple occurrences, a detail page can be added. For example: Outdoor Signs (CP 14 40):



POLICY NUMBER:

COMMERCIAL PROPERTY
CP 14 40 06 07

THIS ENDORSEMENT CHANGES THE POLICY. PLEASE READ IT CAREFULLY.

OUTDOOR SIGNS

This endorsement modifies insurance provided under the following:

BUILDING AND PERSONAL PROPERTY COVERAGE FORM
CONDOMINIUM ASSOCIATION COVERAGE FORM
CONDOMINIUM COMMERCIAL UNIT-OWNERS COVERAGE FORM
STANDARD PROPERTY POLICY

SCHEDULE

Premises Number:		**Building Number:**	
Description Of Sign:			
Construction Of Sign:	☐ Entirely Metal	☐ Other	
Limit Of Insurance:	$		
Coinsurance Percentage:	%		
Additional Premium:	$		
Premises Number:		**Building Number:**	
Description Of Sign:			
Construction Of Sign:	☐ Entirely Metal	☐ Other	
Limit Of Insurance:	$		
Coinsurance Percentage:	%		
Additional Premium:	$		

Name	Parent	Table	Type	Sequence	Sub	Form Type	Number
Outdoor Signs	Building Detail	CommercialPropertyOutsideS	Form	21900	1	Optional	CP 14 40 06 07
Outdoor Signs Summary	Outdoor Signs	CommercialPropertyOutsideS	Summary	21901	1		
Outdoor Signs Detail	Outdoor Signs Summary	CommercialPropertyOutsideS	Detail	21902	1		

Page Outdoor Signs is a form page. Outdoor Signs Detail page is used to capture the schedule on form CP 14 40 shown on the ISO form.

Color Codes

The use of color coding represents the status of the content, whether a page has been added, changed, or deleted from the content compared to the inherited base or parent content:

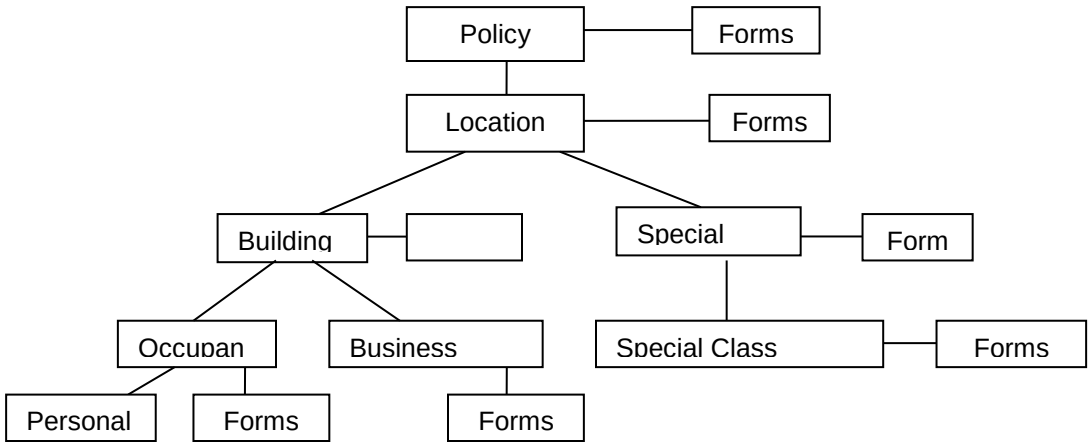
- **Black** - Indicates the content is inherited from the parent content.
- **Blue** - Indicates the content is added or modified from the parent content.
- **Red -** Indicates the content is deleted from the parent content.

How to Use Pages to Generate a Hierarchy or List of Forms for Certain Criteria

The Rating Content pages can be used to create a line of business hierarchy or a list of all the forms available in a policy level. See the steps below followed by an illustration.

Page Hierarchy for Commercial Property:

1. Within the Rating Content file, start from the "Policy Detail" page and work down.
2. Filter type by "Summary", and then select parent "Policy Detail" to identify the policy level risks. Ignore "Payment" page.
3. Repeat the same steps to identify the risks under each of the policy risk levels.

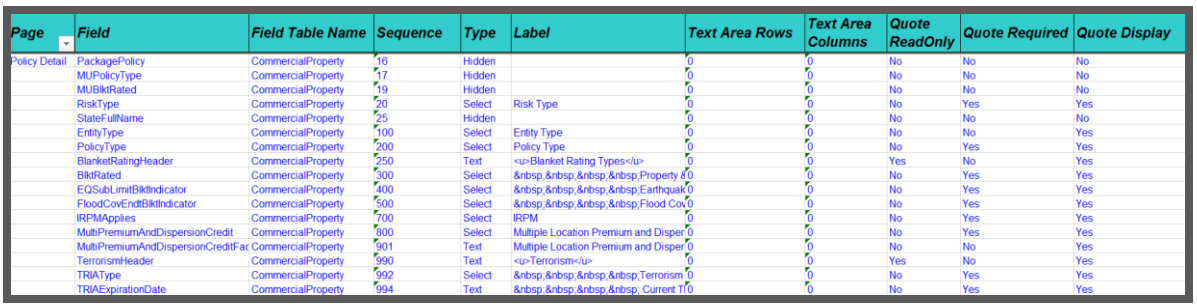


List of forms available in a policy level:

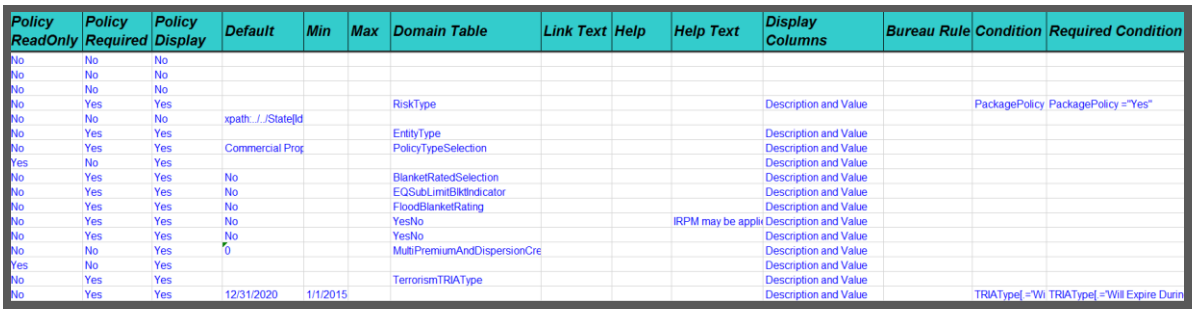
1. Filter type by "Form", and then filter "Parent" by a particular risk level.
2. Conditions can be specified for optional forms. To generate a list of forms for certain criteria, use the conditional filter to obtain a list.

Page Fields Tab

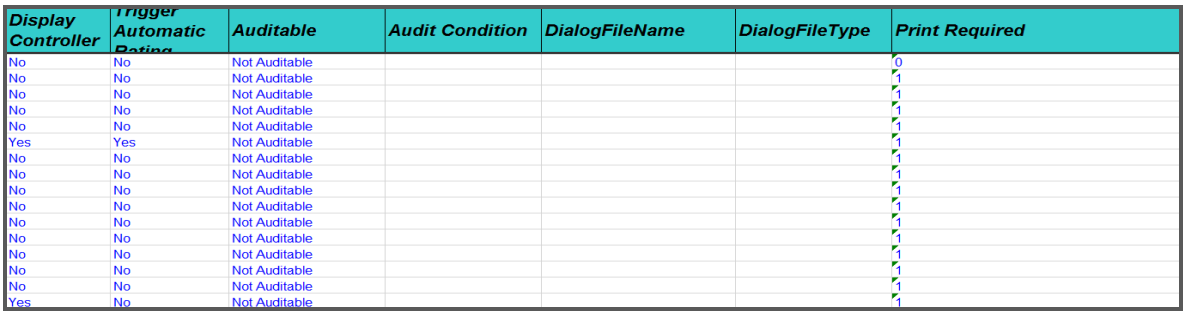
Page Fields are used to capture the information or data required to rate policies. The data captured through the page fields can also be used to determine the form attachment logic. Page fields are always tied to columns. Just as columns are a subset of tables, page fields are a subset of pages. See the example below followed by a brief description of the columns.



Page	Field	Field Table Name	Sequence	Type	Label	Text Area Rows	Text Area Columns	Quote ReadOnly	Quote Required	Quote Display
Policy Detail	PackagePolicy	CommercialProperty	16	Hidden		0	0	No	No	No
	MUPolicyType	CommercialProperty	17	Hidden		0	0	No	No	No
	MUBlktRated	CommercialProperty	19	Hidden		0	0	No	No	No
	RiskType	CommercialProperty	20	Select	Risk Type	0	0	No	Yes	Yes
	StateFullName	CommercialProperty	25	Hidden		0	0	No	No	No
	EntityType	CommercialProperty	100	Select	Entity Type	0	0	No	No	Yes
	PolicyType	CommercialProperty	200	Select	Policy Type	0	0	No	Yes	Yes
	BlanketRatingHeader	CommercialProperty	250	Text	<u>Blanket Rating Types</u>	0	0	Yes	No	Yes
	BlktRated	CommercialProperty	300	Select	Property	0	0	No	Yes	Yes
	EQSubLimitBlktIndicator	CommercialProperty	400	Select	Earthquak	0	0	No	Yes	Yes
	FloodCovEndtBlktIndicator	CommercialProperty	500	Select	Flood Cov	0	0	No	Yes	Yes
	IRPMApplies	CommercialProperty	700	Select	IRPM	0	0	No	Yes	Yes
	MultiPremiumAndDispersionCredit	CommercialProperty	800	Select	Multiple Location Premium and Disper	0	0	No	Yes	Yes
	MultiPremiumAndDispersionCreditFac	CommercialProperty	901	Text	Multiple Location Premium and Disper	0	0	No	No	Yes
	TerrorismHeader	CommercialProperty	990	Text	<u>Terrorism</u>	0	0	Yes	No	Yes
	TRIAType	CommercialProperty	992	Select	Terrorism	0	0	No	Yes	Yes
	TRIAExpirationDate	CommercialProperty	994	Text	Current T	0	0	No	Yes	Yes



Policy ReadOnly	Policy Required	Policy Display	Default	Min	Max	Domain Table	Link Text	Help	Help Text	Display Columns	Bureau Rule	Condition	Required Condition
No	No	No											
No	No	No											
No	No	No											
No	Yes	Yes				RiskType				Description and Value		PackagePolicy	PackagePolicy ="Yes"
No	No	No	xpath:../../State[Id										
No	Yes	Yes				EntityType				Description and Value			
No	Yes	Yes	Commercial Prop			PolicyTypeSelection				Description and Value			
Yes	No	Yes								Description and Value			
No	Yes	Yes	No			BlanketRatedSelection				Description and Value			
No	Yes	Yes	No			EQSubLimitBlktIndicator				Description and Value			
No	Yes	Yes	No			FloodBlanketRating				Description and Value			
No	Yes	Yes	No			YesNo			IRPM may be appli	Description and Value			
No	Yes	Yes	No			YesNo				Description and Value			
No	No	Yes	0			MultiPremiumAndDispersionCre				Description and Value			
Yes	No	Yes								Description and Value			
No	Yes	Yes				TerrorismTRIAType				Description and Value			
No	Yes	Yes	12/31/2020	1/1/2015						Description and Value		TRIAType[='Wi	TRIAType[='Will Expire Durin



Display Controller	Trigger Automatic Rating	Auditable	Audit Condition	DialogFileName	DialogFileType	Print Required
No	No	Not Auditable				0
No	No	Not Auditable				1
No	No	Not Auditable				1
No	No	Not Auditable				1
No	No	Not Auditable				1
Yes	Yes	Not Auditable				1
No	No	Not Auditable				1
No	No	Not Auditable				1
No	No	Not Auditable				1
No	No	Not Auditable				1
No	No	Not Auditable				1
No	No	Not Auditable				1
No	No	Not Auditable				1
No	No	Not Auditable				1
No	No	Not Auditable				1
Yes	No	Not Auditable				1

- **Page -** The name of the page on which the page field is displayed.
- **Field -** The name of the page field. The Page field name is the same as the column name.
- **Sequence -** Determines the order in which the Page fields are displayed on the page.
- **Type -** Controls how data should be entered into the page field -- input data fields versus read-only fields. Page fields can be any of the below type:

- **Anchor** – A hyperlink that the user clicks to invoke a local file, such as an HTML, PDF, or image file. Fields of this type can be ignored.
- **Button** - A search button that the user clicks to perform a search for a client name or address. When using buttons they must be applied to Unique Identifiers that are linked to the Address, Company, or Human table.
- **Checkbox** - A check box where an item can be selected to indicate Yes/No or 1/0
- **Dialog** - A text box and a select button were used. When a user clicks the select button, the system displays an external web page in a web page dialog box. The user can specify the file name of the external web page in the dialog file name field and the type of the file (either HTML or XSL) in the dialog file type field.
- **Hidden** – Is used for system purposes mostly to store information captured at a different risk or policy level which is not required to be displayed. Fields of this type can be ignored.
- **Radio** – Is used to select an item. Radio buttons appear in a horizontal format, and a domain table must be associated with the radio page field type. This is mostly used for Yes/No selections.
- **Radio** - Vertical – Is used to select an item. The radio buttons appear in a vertical format, and a domain table must be associated with the radio page field type.
- **Select** - A drop-down list of items from which the user selects one item. A domain table must be associated with the select page field type.
- **Text** - A text field that can accommodate a single line of user-entered text. Note: Data entered in page fields of this type is controlled by the type of column tied to the page field, for example a page field tied to an integer column cannot have string being entered.
- **Text Area** - A text area can accommodate multiple lines of user-entered text. Text areas are associated with columns of type Text.

- **Label -** Displays the page field label as it will appear in the web page in the user interface.
- **Text Area Rows -** Specifies the number of rows for a text area page field.
- **Text Area Columns -** Specifies the number of columns for a text area page field.
- **Quote Read Only -** This is a Yes/No indicator that specifies that the page field cannot be edited on a quote page. For example, page fields to display premium, rates and factors output from the system.
- **Quote Required -** This is a Yes/No indicator that specifies that the entry is required for the Page Field on a quote page.
- **Quote Display -** This is a Yes/No indicator that specifies that the Page field is displayed on a quote page.

- **Policy Read Only -** This is a Yes/No indicator that specifies that the Page field cannot be edited on a policy page. For example, Page fields to display Premium, Rates and Factors which should not be modified.
- **Policy Required -** This is a Yes/No indicator that specifies that the entry is required on a policy page.
- **Policy Display -** This is a Yes/No indicator that specifies that the Page field is displayed on a policy page.

Note: Different controls exist for quote and policy. If you are working with both quote and policy, these fields may be of value, otherwise ignore them.

Page	Field	Seque	Type	Label	Quote Required	Policy Required
Extra Expense From Dependent Properties Detail	Desc	200	Textarea	Description of Occupancy	No	Yes
	NameID	300	Button	Name	No	Yes
	AddressID	400	Button	Address	No	Yes

Most fields required for premium determination are quote and policy required, however, certain fields used for data capture and statistical assignment purposes may only be policy required. The Occupancy Description, Name and Address fields illustrated in the above example are policy required and not quote required.

- **Default -** Allows the user to enter a default value for the page field. It is also used with anchor type page fields to store the path location for desired document.
- **Min -** This controls the minimum value for a page field.
- **Max -** This controls the maximum value for a page field.
- **Domain Table -** This specifies a domain table that is associated with the page field. The type of fields eligible for domains are radio, radio-vertical, select, and text.

Field	Sequence	Type	Label	Default	Domain Table
BurglaryAlarmSystemType	400	Select	Type of Burglar Alarm System	Not Applicable	BurglaryAlarmSystemType

Domain tables can be associated with a page field to capture information, if there is a list of options. As shown above the “BurglaryAlarmSystemType” domain table is associated with the BurglaryAlarmSystemType page field.

The domain table contains a list of Burglar alarm system types. The Default Parameter contains the default value of the page field if no selection is made.

- **Help -** This allows for the entry of help text to be displayed when the mouse is placed over the page field. This column is for internal purpose and can be ignored.
- **Help Text** - Provides for entry of detailed help text that is displayed when the page field label is clicked. This text is more detailed than the text entered in the “Help” parameter and may exceed 128 characters. This column is for internal purposes and can be ignored.
- **Condition -** This is the condition based on which the page field is displayed.
- **Required Condition -** This is the condition that makes the page field required.

Field	*Condition*	*Required Condition*
BurglaryAlarmCreditFactor	BurglaryAlarmSystemType != "Not Applicable"	BurglaryAlarmSystemType != "Not Applicable"

The Condition parameter specifies the condition for the “BurglaryAlarmCreditFactor” page field to appear in the user interface.

Table	*Column*	*Rating Required*	*Rating Required Condition*
CommercialPropertyBurglaryAndRobberyProtectiveSafeguards	BurglaryAlarmCreditFactor	Yes	BurglaryAlarmSystemType != "Not Applicable"

The Required Condition parameter specifies the condition for the “BurglaryAlarmCreditFactor” page field to be used as a required input for rating.

The column to which the page field is tied also has the rating required condition. This rating required condition can be referenced in the “Ratebook Columns” tab in the ERC file.

- **Display Controller -** This specifies whether the page field controls the display of other page fields on the page. This column is for internal purposes and can be ignored.
- **Print Required –** Identifies variable data fields on the ISO forms that may be required for print. 0 = N or No; 1 = Y or Yes.

The following list of information is only used with the ERC+software product. As of October 2018, this information is being removed from ERC. If you are working with ERC releases prior to this time, please disregard this information.

1. Text Display Size
2. Quote Disabled
3. Policy Disabled
4. Mask, Link Text

5. View Resource
6. Modify Resource
7. Comment
8. Script Before
9. Script After
10. Script Change
11. Amendable Post Issuance
12. Amendable Pre-Renewal
13. XslValue
14. DialogFileName
15. DialogFileType
16. Reference Table
17. LOB Tab Name
18. LOB Tab Sequence
19. LOB Tab Coverage Part
20. Display Mode
21. Display Mode Condition
22. Display Mode Read Only
23. ESI Template File Name
24. ESI Labels Defined In Template

Use of Page Fields

A page field can be data collect or data display. To capture all data collect fields, filter "Quote Required" or "Policy Required" fields by "Yes" and also ignore all read-only fields.

How to Identify Coverages

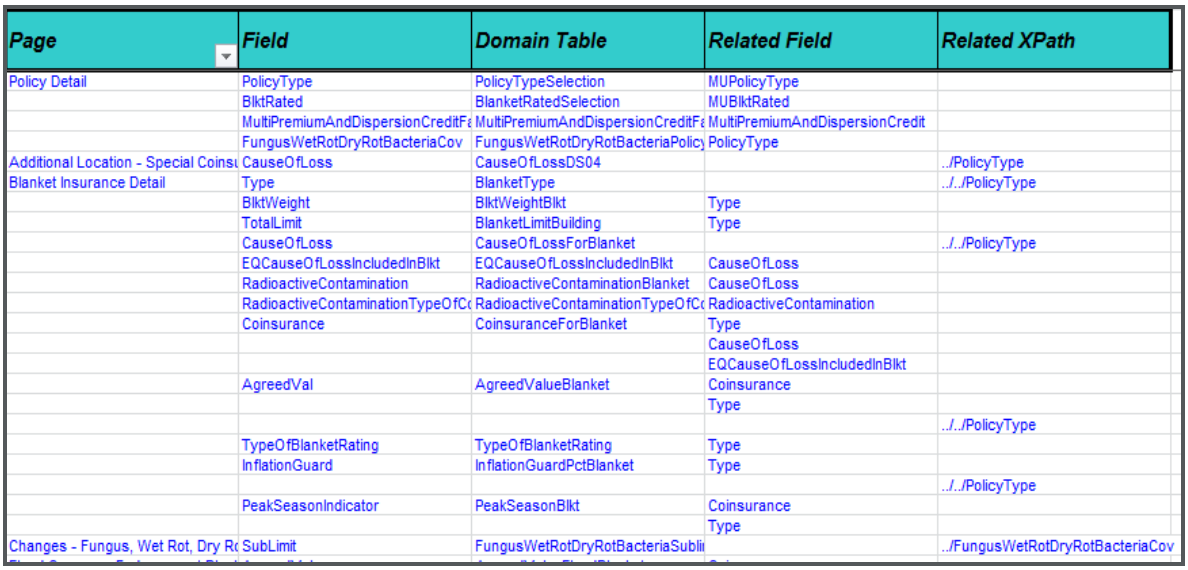
To identify coverages for a line of business, identify the page fields with the format Table.Column.
For example:
CommercialPropertyPersonalPropertyPrsnlPropCoverage.PolicyTermPremium, this identifies coverage CommercialPropertyPersonalPropertyPrsnlPropCoverage.

How to Identify Data Capturing Fields for a Form

To identify data capturing fields on a form, select the page name of the form and generate the fields on the form page or detail page.

Page Related Fields Tab

The purpose of the Page Related Fields tab is to identify the controlling elements for a particular domain table. This will structure any page field that has a domain table associated with it. If a domain table has a key or pre-requisite values associated with it, the controlling elements need to be aligned with it.



Page	Field	Domain Table	Related Field	Related XPath
Policy Detail	PolicyType	PolicyTypeSelection	MUPolicyType	
	BlktRated	BlanketRatedSelection	MUBlktRated	
	MultiPremiumAndDispersionCreditFa	MultiPremiumAndDispersionCreditFa	MultiPremiumAndDispersionCredit	
	FungusWetRotDryRotBacteriaCov	FungusWetRotDryRotBacteriaPolicy	PolicyType	
Additional Location - Special Coinsu	CauseOfLoss	CauseOfLossDS04		../PolicyType
Blanket Insurance Detail	Type	BlanketType		../../PolicyType
	BlktWeight	BlktWeightBlkt	Type	
	TotalLimit	BlanketLimitBuilding	Type	
	CauseOfLoss	CauseOfLossForBlanket		../../PolicyType
	EQCauseOfLossIncludedInBlkt	EQCauseOfLossIncludedInBlkt	CauseOfLoss	
	RadioactiveContamination	RadioactiveContaminationBlanket	CauseOfLoss	
	RadioactiveContaminationTypeOfC	RadioactiveContaminationTypeOfC	RadioactiveContamination	
	Coinsurance	CoinsuranceForBlanket	Type	
			CauseOfLoss	
			EQCauseOfLossIncludedInBlkt	
	AgreedVal	AgreedValueBlanket	Coinsurance	
			Type	
				../../PolicyType
	TypeOfBlanketRating	TypeOfBlanketRating	Type	
	InflationGuard	InflationGuardPctBlanket	Type	
				../../PolicyType
	PeakSeasonIndicator	PeakSeasonBlkt	Coinsurance	
			Type	
Changes - Fungus, Wet Rot, Dry Ro	SubLimit	FungusWetRotDryRotBacteriaSubli		../FungusWetRotDryRotBacteriaCov

The way to read the information in this tab is as follows:

- **Page** – This column identifies the page where the page field resides (e.g., Policy Detail).
- **Field** - Provides the name of the page field that has a domain table associated with it.
- **Domain Table** – Identifies the name of the domain table associated with that field.
- **Related Field** and **Related XPath** - Are mutually exclusive; you can't have values in both.

Algorithm (ALG) Files

The Algorithm files in ERC outline business rules from the ISO CLM for a particular line of business that are in effect at a certain point in time, including but not limited to business rules for:

- Form attachment logic for endorsements/forms
- Statistical code assignment logic
- Rating messages
- Premium development and rating procedures

Structure of Algorithm (MS Excel version)

There are many worksheets or tabs in an algorithm (ALG) file, the number of which vary by line of business. However, all ALG files start with a Table of Contents and Application tab and end with a Flowchart tab; in between are separate tabs for each algorithm. See a summary of ALG tabs below.

- **Table of Contents** - Provides a list of, and links to, all the algorithms contained in the ALG file. See example below followed by a brief description of each column.



Algorithm	Rule Reference	StateIndicator
Policy Common Rating	Common Rating	CW
Error Messages	ErrorMessages	CW
Conditional Mandatory Form Logic	Form Logic	CW
Coverage Indicator	policyIndicator	CW
Stat Assignment Info	Stat Assignment Info	CW
Err-CommercialPropertyLocation	Err-Location	CW
Err-CommercialPropertySpecialClass	Err-SpecialClass	CW
Err-CommercialPropertyAdditionalPropertyNotCoveredSpecialClass	Err-AdditionalPropertyNotCov	CW
Err-CommercialPropertyDeductiblesByLocationSpecialClass	Err-DeductiblesByLocationSp	CW
Err-CommercialPropertySpecialClassBusnIncome	Err-SpecialClassBusnIncome	CW
Err-CommercialPropertySeasonalLeasesMonthlyLimitsOnLossPaymentSpclClassBusnIncome	Err-SeasonalLeasesMonthlyL	CW
Err-CommercialPropertySpecialClassBusnIncomeInterruptionOfComputerOperationsCoverage	Err-SpecialClassBusnIncomeI	CW
Err-CommercialPropertyStructure	Err-Structure	CW
Err-CommercialPropertyAdditionalPropertyNotCovered	Err-AdditionalPropertyNotCov	CW
Err-CommercialPropertyBusinessIncome	Err-BusinessIncome	CW
Err-CommercialPropertyBusinessIncomeInterruptionOfComputerOperationsCoverage	Err-BusinessIncomeInterrupt	CW

  - **Algorithm** - Lists the worksheet pages in the order that they appear in the ALG file.

    - **Rule Reference** - Identifies the ISO CLM rule number associated with the calculation. If a rule does not apply, then other descriptive information is provided.
    - **State Indicator** - Indicates whether the primary calculation header (PCH) within the ALG is countrywide (CW) or state specific (e.g., AK).
- **Application Version** – Identifies the version and date of the parser used to generate the ALG.
- **Error Messages** – Provides the logic for all validation messages.
- **Stat Assignment Info** – Provides the logic for all statistical code assignment.
- **Form Logic** – Provides the logic for form attachment logic.
- **Common Rating or Risk Common Rating** - Contains common rating elements for either the policy or for a given risk.
- **Total Policy Premium or Total Risk Premium** - Outlines the accumulation of premiums for all child coverages, forms, and risk for a given level
- **Coverage on Policy Indicator** - Contains the conditions under which a coverage is included on a policy.
- **Flowchart** – Outlines in a tree structure the exact order of processes that should be followed in order to calculate a premium correctly.

The following are available for each risk, coverage or form that is premium bearing.

- **Table Objects** - Provides a list of all Table nodes.
    - Illustrates a hierarchy review.
    - Illustrates which level (State or CW) the table applies to.
- **Column Nodes** – Provides a list of all Columns nodes. Each is identified as “Used” or/and “Used P”.
    - **Used** – Column Node is used in Rating but not in a Premium Calculation on that table.
    - **Used P** – Column Node is used in Rating and in the Premium Calculation on that table.

  To determine all column definitions for those active values, a compiled list can be created using this tab. Once compiled, this list can be used as a cross reference with the DS file under “Column Details” to pull out the exact definition of each object/column, including type and size.

Algorithm Body and Operations

Below is an example of an algorithm followed by a description of its contents.

BOF		Rule#	Name	EOL
RUL	Primary Rule	Rule 41.E	Commercial Property Builders Risk Theft Of Building Materials Fixtures Machinery Equipment.Premium	EOL
TAB	DBTablesUsed		Commercial Property Builders Risk Theft Of Building Materials Fixtures Machinery Equipment	EOL
BBB	Total Premium		Commercial Property Builders Risk Theft Of Building Materials Fixtures Machinery Equipment	EOL
COM	Type	Operator	Value	EOL
PCH			Premium	EOL
CVS	Assignment		Premium	EOL
CVM		=		EOL
CVM		(		EOL
CVM		(		EOL
CVM		(		EOL
CVM		(	Factor	EOL
CVM		*	Commercial Property Structure \ Basic Group I Rate	EOL
CVM		) Round Up Thousandth		EOL
CVM		*	Limit	EOL
CVM		/	100.0	EOL
CVM		) Round Up Dollar		EOL
CVM		*	Commercial Property \ Package Mod Factor	EOL
CVM		) Round Up Dollar		EOL
CVM		*	Commercial Property \ IRPM Factor	EOL
CVE		) Round Up Dollar		EOL
PCH			Factor	EOL
CVS	Assignment		Factor	EOL
CVE		=	Rate Table: Builders Risk Theft Of Building Materials Fixtures Machinery Equipment Factor(Using "Y")	EOL
PCH			Premium Indicator	EOL
IFB	Condition	IF		EOL
IFM		(		EOL
IFM			Premium	EOL
IFM		Not Equal To	0	EOL
IFE		)		EOL
CVS	Assignment		Premium Indicator	EOL
CVE		=	1	EOL
ELS		ELSE		EOL
CVS	Assignment		Premium Indicator	EOL
CVE		=	0	EOL
EIF		END IF		EOL
EOF	TYP	OPR	VAL	

- **Primary Rule** – Identifies the rule number from the ISO CLM and the name of the table object.
- **DB Tables Used** – Identifies the name of the database table used.
- Calculation headers are highlighted in purple indicating a box of logic for the assignment. The box of logic includes assignments and conditions. The premium calculation is first established and the logic then works its way down to get each individual rate and factor.
- **Column A** – Is used to facilitate machine readability. See acronym definitions below:

Acronym	Description
BBB	Begin Block Build
BOF	Beginning of File
CCH	Class Calculation Header
COM	Comment
CVE	Calculation End
CVM	Calculation Body
CVS	Calculation Start
EIF	End IF

ELS	Else
EOF	End of File
FRS	FOR LOOP Start
IFB	IF Begin
IFE	IF End
IFM	IF Body
PCH	Premium Calculation Header
RUL	Rule Information
SCH	Set node Calculation Header code
TAB	DB Table used

- **Column B** – Provides an indicator that defines code as either CW or State. See acronym definitions below:

Acronym	Description
CW1/CW2	CountryWide
SRC	Source
ST1	State

- **Column C** - Defines the three types of logic used in an algorithm:
  1. **Assignment** – Identifies the start of the assignment name
  2. **Condition** – Identify if/else statements
  3. **Looping** - Indicates a looping function
- **Column D** – Provides the types of Operators. The list of operators includes but is not limited to the following:
  - **Numerical Operands**: - =, +, -, *, /
  - **Logical Operands:** Or, And, Is Equal To
  - **IF/ELSE**
  - **Rounding**
- **Column E** - Defines the variables
- **Column F** - Defines the variable Type

Acronym	Description
COLUMN_NUMERIC	Class Column Nodes
COLUMN_STRING	Class Column Nodes - String
CONSTANT	Constant
LV_BOOLEAN	Local Variable of type boolean
LV_DATE	Local Variable of type date
LV_DOUBLE	Local Variable of type double
LV_INTEGER	Local Variable of type integer
LV_STRING	Local Variable of type string
LV_TimeSpan	Local Variable that stores the difference between dates in an integer format. This difference can be Days, Months, or Years
RT	Rate Table
UI	User Input Field
XPATH	Other Class Column Nodes
XPATH_NUMERIC	Other Class Column Nodes - Numeric
XPATH_STRING	Other Class Column Nodes – String

- **Column G** - Defines the end of logic

The following table contains a list of XML elements including their attributes and a parent/child relationship:

Element	Attributes	Possible Child Elements
Arg	number variableType	
assign	state recordType lValue variableType	expression
ATTACH	state recordType form	
bracket	expression	
class	name type pass	class rating stat err formAttachment premium

ColumnNode	usage	
condition	state recordType	expression
Else	state recordType excelSheet excelRow	if else assign loop
Err	className variables type	
expression	side ruleReference recordType Op asText variableType	expression function ratetable bracket
formAttachment	className variables type	
function	Type	arg
if	state recordType excel Sheet excel Row	condition then else ATTACH OptOutForm
LOB	name RatebookId Version ParentName AssemblyId	reference
loop	through state recordType	if loop assign
OptOutForm	OptOut Key FormNumber FormName	
PCH	state number name recordType excelSheet excel Row overridden	if then else assign loop class ATTACH OptOutForm

Premium	className variables type pass	
ratetable	code	arg
Rating	className variables type pass	
reference	number name Type dbTables state inherited	class scope TableName
scope	ruleReference recordType dbTables pass state inherited	if PCH assign loop class ATTACH
Stat	className variables type	
TableName	name ruleReference multipleAllowed hasPremium foundInLayer	ColumnNode TableName
then		assign if loop ATTACH OptOutForm

The following table provides the purpose of each element within the XML.

Element	Purpose
arg	Defines the values that are being used within either a rate table look up or a count function
assign	Defines the process when a value is being set
ATTACH	Associated with conditional mandatory logic and is used to illustrate the process of attaching a form
Bracket	Used to demonstrate order of operations within the code

class	These are tied to the execution flow reference type and are used to identify the table being called in the order of operations for the ALG
ColumnNode	Defines every column on a table (same as in the DS file) and determines if it is used within the rating process
condition	The condition element is a child of an if node and is the starting point of the body of the condition
Else	If the condition for an If statement is false this else contains the catch
Err	Tells which error message PCH's should be rated within the Err-table name reference
expression	An expression defines logic within the ALG, whether it is a comparison or an assignment
formAttachment	Tells which form attachment PCH's should be rated within the form-table name reference
function	Only available function is a count where tally logic is applied. The arg listed would be the table being tallied up
if	Starts the section of XML tied to a condition within the logic
LOB	Starting Element of the XML structure. Identifies the Line Of Business, Ratebook ID and the Version number of the parser used to create the ALG
loop	Defines the process of looping through the Line of Business hierarchy. The path is provided in the through attribute and the starting point is the scope of the reference
OptOutForm	Tied to General Liability Opt out Functionality
PCH	Defines either a column node or a table that is being set within the proceeding XML
Premium	Tells when a table should generate Premium and what the associated PCH's associated with that calculation
ratetable	Defines the call to do a rate table lookup using the Args provided as the criteria elements
Rating	Tells which PCH's should be rated within the reference for the tables pass in the ALG Flow Chart
reference	Contains the section name of the ALG. The name is tied to the tab names within the Excel version. The reference also states the purpose of the section, whether it is an error message, premium calculation or common rating
scope	Defines the table associated with the logic within the reference
Stat	Tells which stat assignment PCH's should be rated within the Stat Assignment reference

TableName	Defines the tables included within the ALG, provides the Rule Reference and displays if table is associated with a Premium Calculation
Then	Then elements are associated with If elements and contain the action when the condition is true

The following table provides the purpose of each attribute within the XML:

Attribute	Purpose
AssemblyId	This stores a GUID that will allow a connection between the ALG and the Id value stored in the RC XML
asText	Give a definition of what the intention is of the following section of XML, this is for documentation purposes only and does not impact the rating process
className	Contains the name of the table where logic is being applied
code	Contains the name of the Rate Table being used in the Rating Process
dbTables	Defines the table used within a reference
excelRow	Documents the row number within the excel version that contains the logic for the following section of XML. This is used for documentation only and does not impact the rating process
excelSheet	Documents the sheet tab within the excel version that contains the logic for the following section of XML. This is used for documentation only and does not impact the rating process
form	Defines the table being attached within an attach element
FormName	Used for Opt Out functionality within GL
FormNumber	Used for Opt Out functionality within GL
foundInLayer	Defines where table is found within the different hierarchy layers
hasPremium	Defines if a table has a premium calculation
Inherited	This is an attribute on references that shows is logic is deviated from CW or not
Key	Used for Opt Out functionality within GL
lValue	Assignment logic where value is being assigned is on the left side of the equation
multipleAllowed	Defines if a table is allowed multiple occurrences or not, if so the RC file gives the maximum value

name	Defines the name of the object for a PCH element
number	Defines order to the element Arg
Op	Defines the operator for an expression
OptOut	Used for Opt Out functionality within GL
Overridden	Shows where value is inherited from CW or deviated within the state. It also shows if state specific logic is unique to the state or a deviation from CW
ParentName	With the state file it gives the name of the CW the ALG inherits from
pass	Gives an order when references are called multiple times, pass references to the order call for references
RatebookId	Defines the name of the ALG
Rating	Defines variables for class files that are used in rating the ALG but are not directly tied to a premium calculation
recordType	Contains the acronym for machine automation purposes on what the element is used for in the ALG
ruleReference	Defines what rule the table is associated with
side	Defines which side of an express the value applies
state	Denotes whether content is CW versus State specific
through	Defines the X Path logic for a looping statement
Type	Defines the type of reference
usage	Determines whether column node is used within the ALG or Not
Variables	An attribute of the Rating element, it list which PCH's are to be rated within that pass in the Flowchart
variableType	Gives a definition of what the value is in an expression
Version	Defines the version number of the parser used to generate the ALG

The following table provides a description of the operator values common in every line of business:

Operator Value	Description
-	Subtraction
!=	Not Equal To
%	"Mod" function
&	AND
&&	AND
*	Multiply
/	Divide
||	OR
+	Addition
<	Less than
<=	Less than or Equal To
==	Equal To (comparison not assignment)
>	Greater Than
>=	Greater than or Equal To
AddYears	Increment the year place holder in a date format (dd/mm/yyyy)
Ceiling	Converts a decimal to the next whole integer For example, 7.03 becomes 8
DateTime.IsLeapYearEffDate	Determines if the time frame provided falls within a leap year
Double.ParseLimit	Interpolation command that determines rate rounded to the thousandth
Double.ParselowerDataRow["Factor"]	Interpolation command that determines rate rounded to the thousandth, value grabs lower factor
Double.ParselowerDataRow["Limit"]	Interpolation command that determines rate rounded to the thousandth, value grabs lower limit
Double.ParseupperDataRow["Factor"]	Interpolation command that determines rate rounded to the thousandth, value grabs upper factor
Double.ParseupperDataRow["Limit"]	Interpolation command that determines rate rounded to the thousandth, value grabs upper limit
IndexOf	Used only in General Liability Opt Out Logic and is used to parse an array list
Key***xxxxx.xx***FormNumber***C***	Used in General Liability Opt Out tab to store Key

G xx xxFormName***../../"Database TableName"***OptOut***Boolean***	***(class code, xxxxx.xx)***, Form Number***(CG xx xx)***, Form Name ***(database table and xpath)*** and Opt Out Choice **(boolean expression)** to an Array List
KeyoptOutClassCode	Used in General Liability Opt Out tab to indicate key within an array list
LeapTest.AddDays1.0	Leap Year Test for Time Range
Math.RoundARDSpan.Days	Rounds a Time Span in Days
Node.OwnerDocument.DocumentElement.Attributes.GetNamedItem"ratingIteration"	Grabs the value of the attribute ratingIteration from the policy XML
Not	Not True
Replace	This replaces a value with another
Round Up Dollar	Round to the dollar
Round Up Thousands	Round to the thousandths
StartsWith	StartsWith, this command
Subtract	Used in data comparison to generate a time range by days
With	A function applied with Replace, Replace a value "With" a new value

The following table provides the purpose of the functions that can be applied to values in the ALG.

Function	Purpose
.Date	Returns the date stored in a local variable
	For example
	ArdDate.Date
.Days	Applied in conjunction with "TimeSpan" variables and allows dates to be compared using days as the basis
	For example
	TimeSpan varaible = date1.Days - date2.Days
.length	Returns the length of the string value, value is an integer
.Month	Returns the month portion of a date;
	For example
	Date - 1/1/2015
	Date.Month = 1
.Months	Applied in conjunction with "TimeSpan" variables and allows dates to be compared using months as the basis
	For example
	TimeSpan variable = date1.Months - date2.Months
.PadLeft	Used with columns or local variables that are strings and allows the stored value to equal a certain amount of characters
	For example
	String - 20, if PadLeft is applied with a value of 3
	String becomes 020
.Value	Returns the values that are stored in a column
.Year	Returns the year portion of a date;
	For example
	Date - 1/1/2015
	Date.Year = 2015
.Years	Applied in conjunction with "TimeSpan" variables and allows dates to be compared using years as the basis
	For example
	TimeSpan varaible = date1.Years - date2.Years

The following table provides a description of the types of elements that can be applied to values in the ALG.

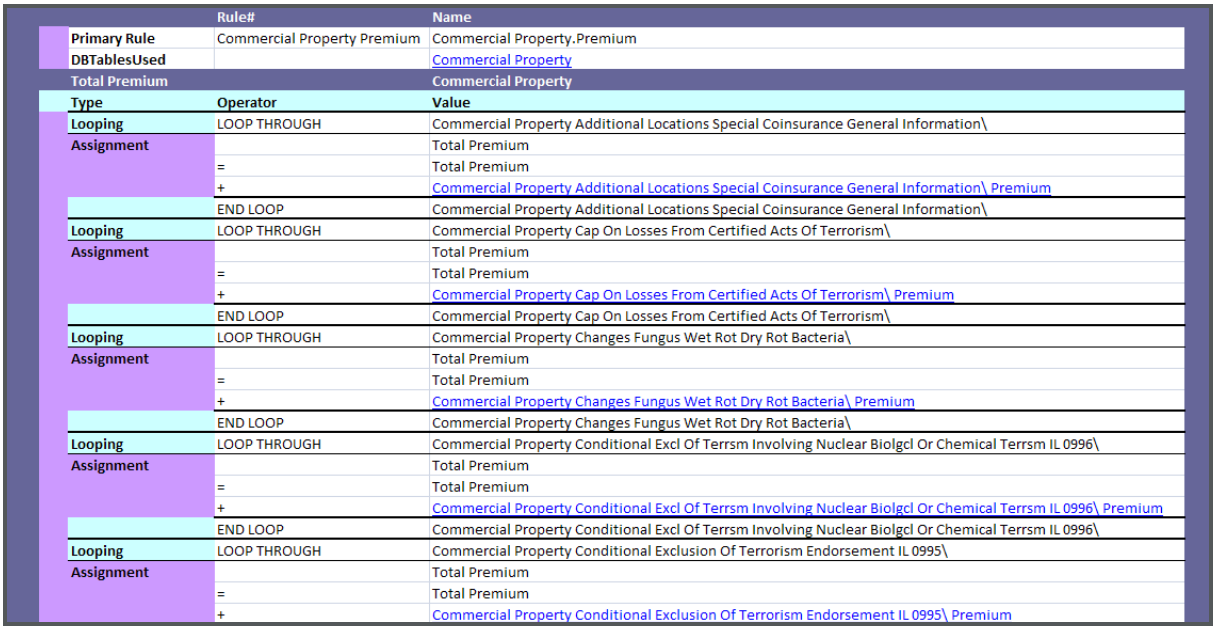
Type	Description
Common Rating	Reference element contains rating logic not tied to a premium calculation
Error Messages	Reference element contains underwriting error message logic for the provided dbTable
Execution Flow	Reference element contains a order of operation flow for the ALG
Form Logic	Reference element contains conditional mandatory logic and mandatory logic for the provided dbTable
Opt Out	Reference element contains the Opt Out logic, only for General Liability
Policy Indicator	Reference element contains the coverage on policy indicator logic for all coverages
Premium Calculation	Reference element contains a premium calculation
Stat Reporting	Reference element contains the statistical assignment logic for all stat codes
Table Objects	Reference element contains the definitions of the table objects and column nodes

The following chart provides a description of the usage types used in the ALG.

Usage	Description
BOTH	Column Node is used in Rating and in the Premium Calculation on that table
OTHERS	Column Node is used in Rating but not in a Premium Calculation on that table
UNUSED	Column Node is not used within the ALG

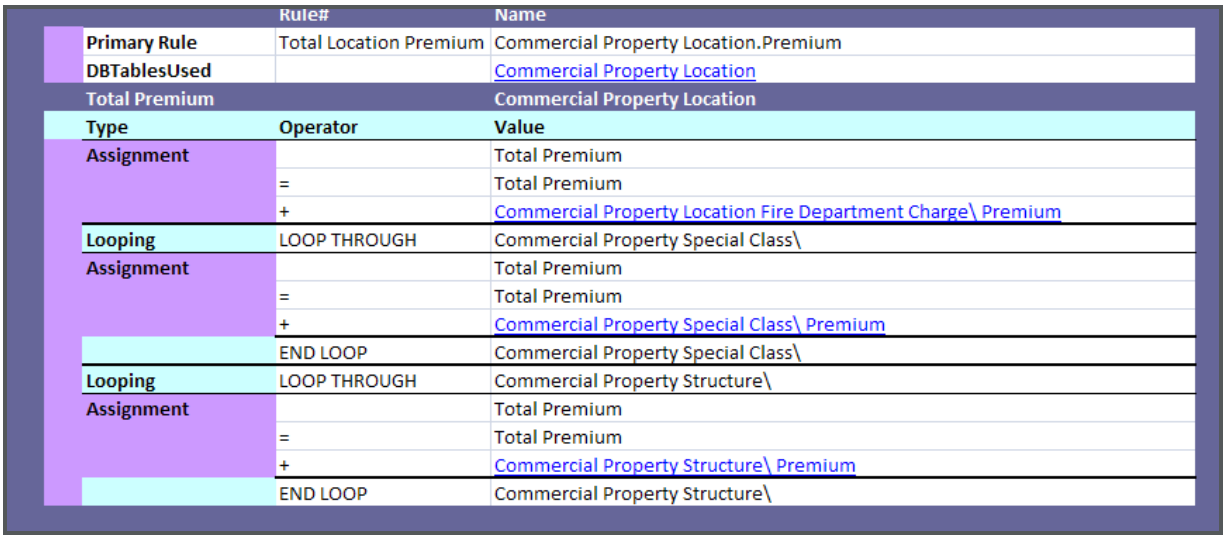
Navigation Example

The total Policy Premium tab is a guide which follows the database data structure. Every risk or form is linked inside a tab to navigate to the corresponding tabs.



	Rule#	Name
Primary Rule	Commercial Property Premium	Commercial Property.Premium
DBTablesUsed		Commercial Property
Total Premium		Commercial Property
Type	Operator	Value
Looping	LOOP THROUGH	Commercial Property Additional Locations Special Coinsurance General Information\
Assignment		Total Premium
	=	Total Premium
	+	Commercial Property Additional Locations Special Coinsurance General Information\ Premium
	END LOOP	Commercial Property Additional Locations Special Coinsurance General Information\
Looping	LOOP THROUGH	Commercial Property Cap On Losses From Certified Acts Of Terrorism\
Assignment		Total Premium
	=	Total Premium
	+	Commercial Property Cap On Losses From Certified Acts Of Terrorism\ Premium
	END LOOP	Commercial Property Cap On Losses From Certified Acts Of Terrorism\
Looping	LOOP THROUGH	Commercial Property Changes Fungus Wet Rot Dry Rot Bacteria\
Assignment		Total Premium
	=	Total Premium
	+	Commercial Property Changes Fungus Wet Rot Dry Rot Bacteria\ Premium
	END LOOP	Commercial Property Changes Fungus Wet Rot Dry Rot Bacteria\
Looping	LOOP THROUGH	Commercial Property Conditional Excl Of Terrsm Involving Nuclear Biolgcl Or Chemical Terrsm IL 0996\
Assignment		Total Premium
	=	Total Premium
	+	Commercial Property Conditional Excl Of Terrsm Involving Nuclear Biolgcl Or Chemical Terrsm IL 0996\ Premium
	END LOOP	Commercial Property Conditional Excl Of Terrsm Involving Nuclear Biolgcl Or Chemical Terrsm IL 0996\
Looping	LOOP THROUGH	Commercial Property Conditional Exclusion Of Terrorism Endorsement IL 0995\
Assignment		Total Premium
	=	Total Premium
	+	Commercial Property Conditional Exclusion Of Terrorism Endorsement IL 0995\ Premium

Following is the link to the Location Premium tab.



	Rule#	Name
Primary Rule	Total Location Premium	Commercial Property Location.Premium
DBTablesUsed		Commercial Property Location
Total Premium		Commercial Property Location
Type	Operator	Value
Assignment		Total Premium
	=	Total Premium
	+	Commercial Property Location Fire Department Charge\ Premium
Looping	LOOP THROUGH	Commercial Property Special Class\
Assignment		Total Premium
	=	Total Premium
	+	Commercial Property Special Class\ Premium
	END LOOP	Commercial Property Special Class\
Looping	LOOP THROUGH	Commercial Property Structure\
Assignment		Total Premium
	=	Total Premium
	+	Commercial Property Structure\ Premium
	END LOOP	Commercial Property Structure\

Below is the Building Basic Group I Coverage tab. It first develops the calculation for premium and then works its way down to develop the rate(s) and factor(s) that are involved.

	Rule#	Name
Primary Rule	Rule 70 & 71 & 72	Commercial Property Structure Building Basic Group I Coverage.Premium
DBTablesUsed		Commercial Property Structure Building Basic Group I Coverage
Total Premium		Commercial Property Structure Building Basic Group I Coverage
Type	Operator	Value
		Premium
Assignment		Premium
	=	Calc Premium
		Calc Premium
Assignment		Calc Premium
	=	0
Condition	IF	
	(	
		Commercial Property Structure\Commercial Property Builders Risk Reporting Form
	Not Equal To	Null
	And	Commercial Property Structure\Cov Type
	Is Equal To	"Builders Risk"
	)	
Assignment		Calc Premium
	=	0
	END IF	
Condition	IF	
	(	
		Commercial Property Structure\Commercial Property Builders Risk Reporting Form
	Is Equal To	Null
	And	Commercial Property Structure\Included In Blkt
	Is Equal To	"No"
Condition	IF	
	(	
		Commercial Property Structure\Commercial Property Builders Risk Reporting Form
	Is Equal To	Null
	And	Commercial Property Structure\Included In Blkt
	Is Equal To	"No"
	And	
	(	Commercial Property Structure\Cov Type
	Is Equal To	"Builders Risk"
	Or	Commercial Property Structure\Cov Type
	Is Equal To	"Building"
	Or	Commercial Property Structure\Cov Type
	Is Equal To	"Improvements and Betterments"
	Or	Commercial Property Structure\Cov Type
	Is Equal To	"Condominium Association"
	)	
	)	
Assignment		Calc Premium
	=	
	(	Commercial Property\IRPM Factor
	*	
	(	Commercial Property Structure\Multi Premium And Dispersion Credit Factor
	*	
	(	Commercial Property Structure\Package Mod Factor
	*	
	(	Commercial Property Structure\Final Basic Group I Rate
	*	Commercial Property Structure\Limit
	/	100.0
	) Round Up Dollar	
	) Round Up Dollar	
	) Round Up Dollar	
	) Round Up Dollar	
	END IF	

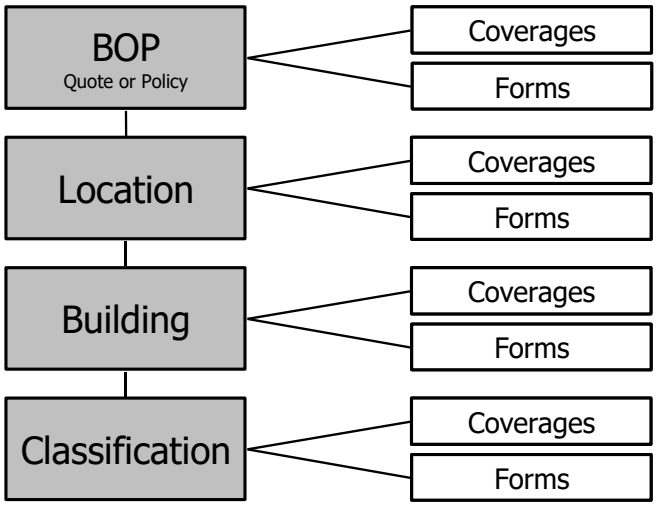
In the first example, it shows that based on the Cov Type, premium = Final Basic Group Rate x Limit/1000 x IRPM Factors x Multi Premium And Dispersion Credit Factor X Package Mod Factor.

Policy Administration Elements

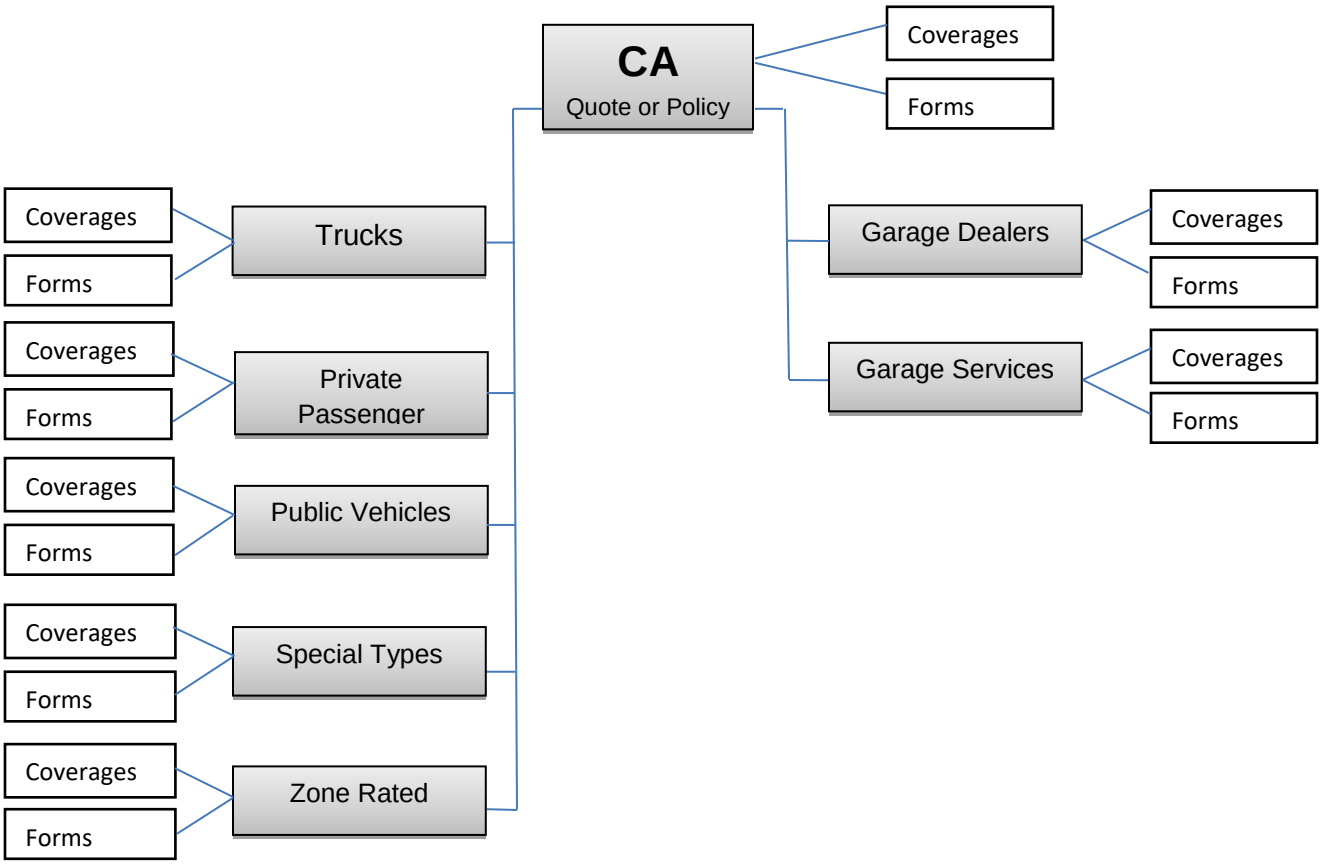
The values listed below will not be found within the rating content for any lines of business. However, they may be referenced in the algorithm file for some lines. Each of these values is a sibling of the main line of business table element in the XML structure for a line of business.

- **AnnDate** - Anniversary Rating Date for Workers Comp line of business
- **EffDate** - Effective date of the quote/policy being rated
- **ExpDate** - Expiration date of the quote/policy being rated
- **Renewal** - Indicates whether or not the policy is a renewal policy or a first issuance.
- **RenewalIteration** – Identifies the number of times a policy has been renewed.
- **Transaction/Effective** – Identifies the date of a modification made to a policy. This can include creation, issue, or endorsement.
- **State/Name** - Contains the full name of the State that is being rated.
- **State/ID** - Contains the unique identifier for that state.
- **StateId** - Contains the ID of the state associated with the RC.

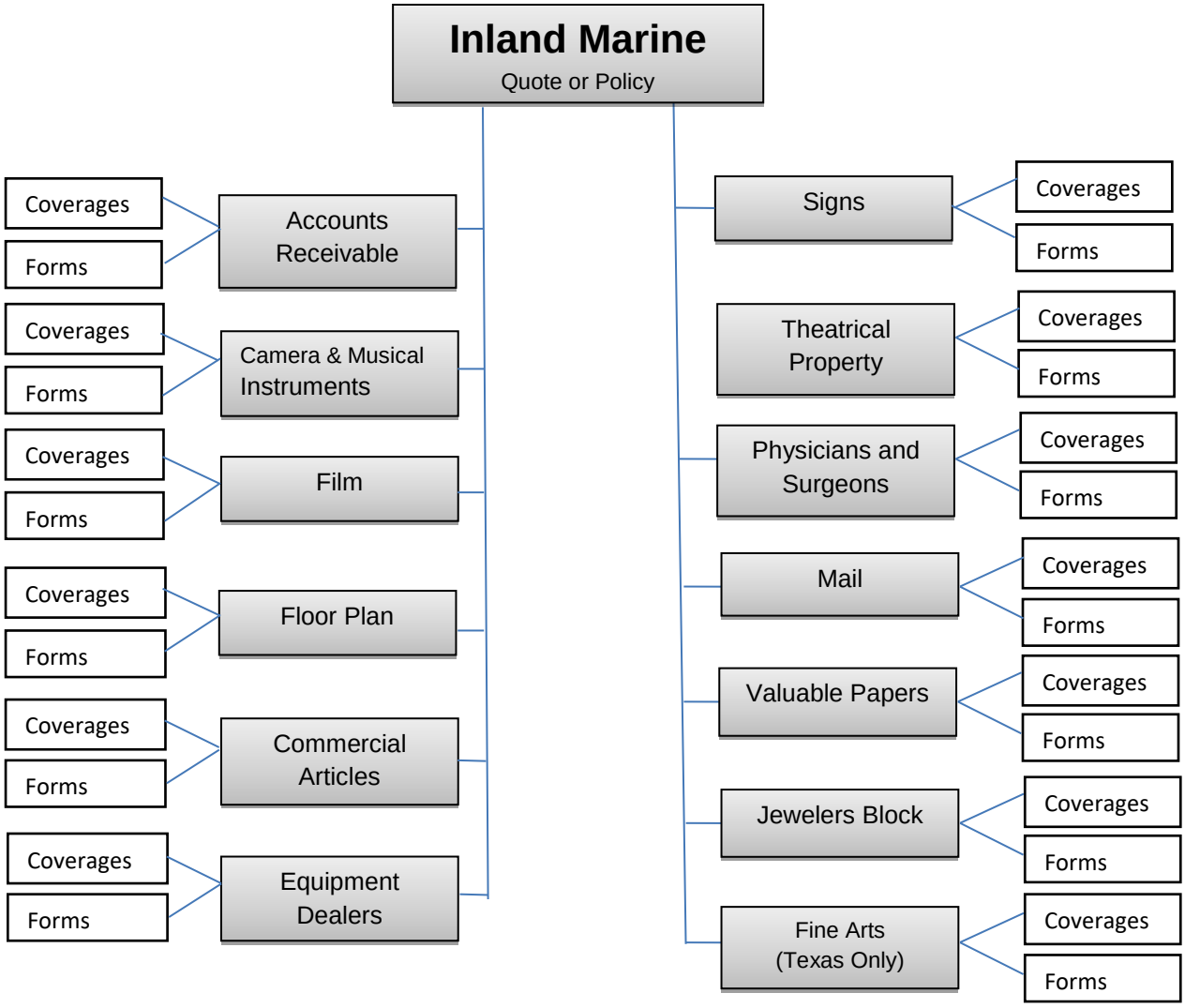
Businessowners Hierarchy



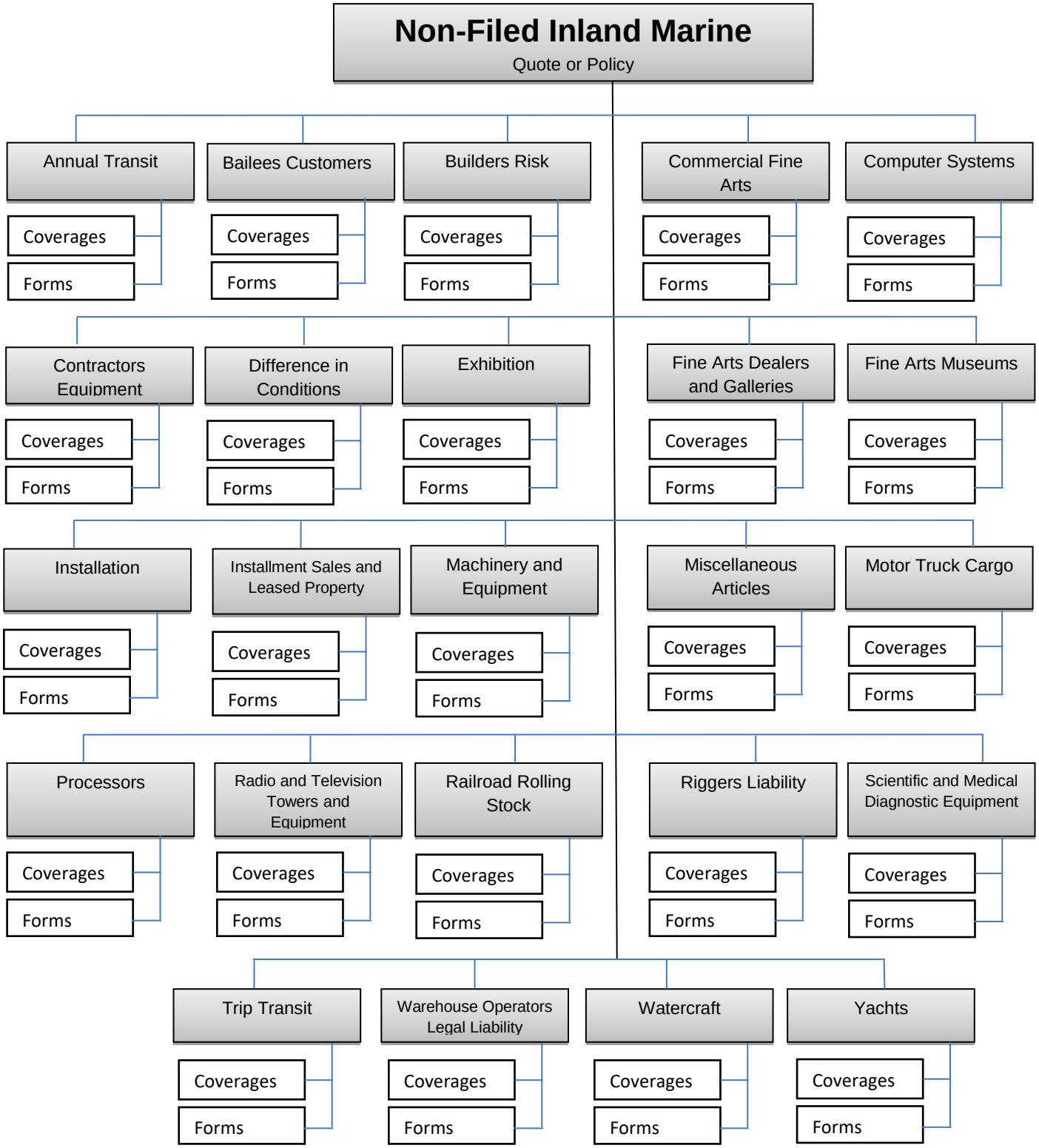
Commercial Automobile Hierarchy



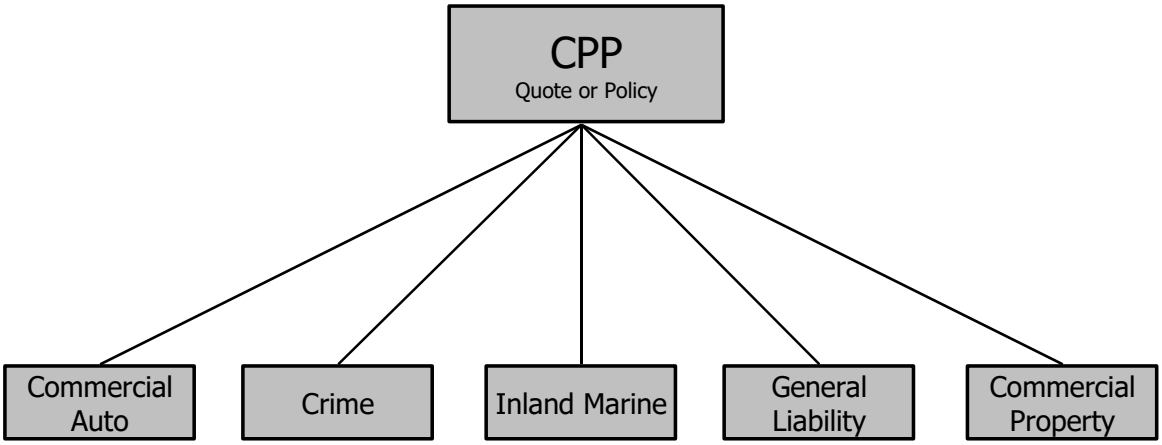
Commercial Inland Marine (Filed) Hierarchy



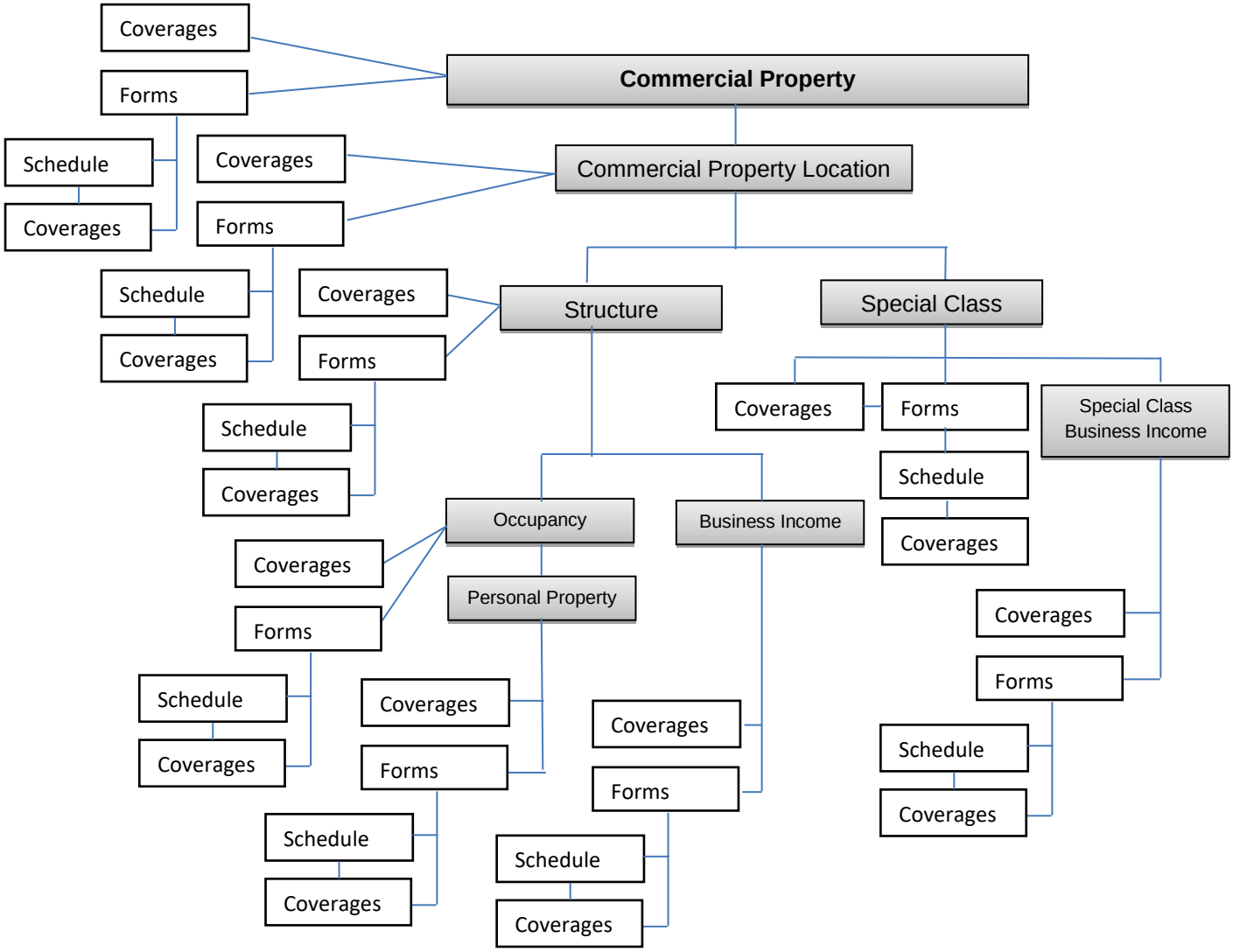
Commercial Inland Marine (Non-Filed) Hierarchy



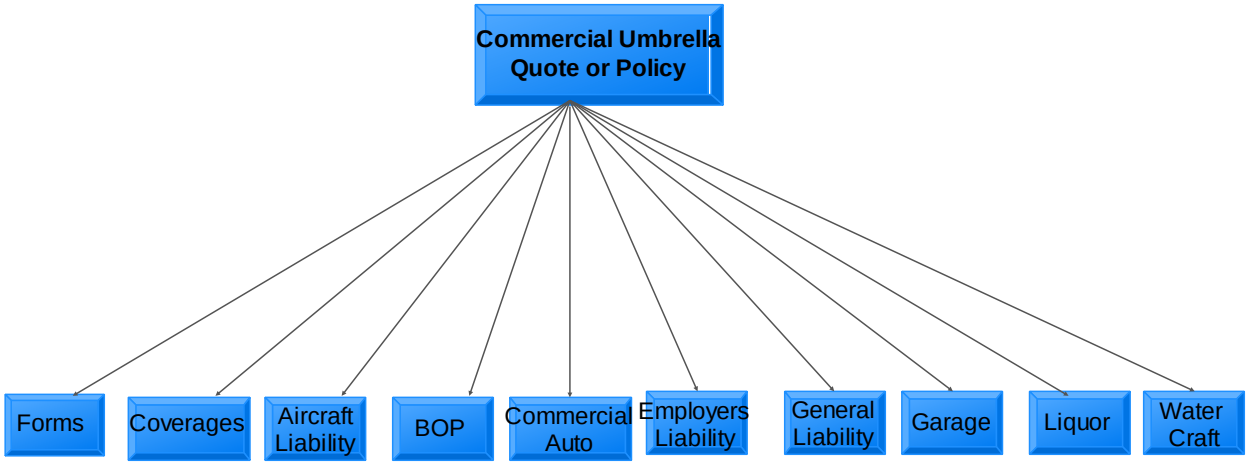
Commercial Package Policy Hierarchy



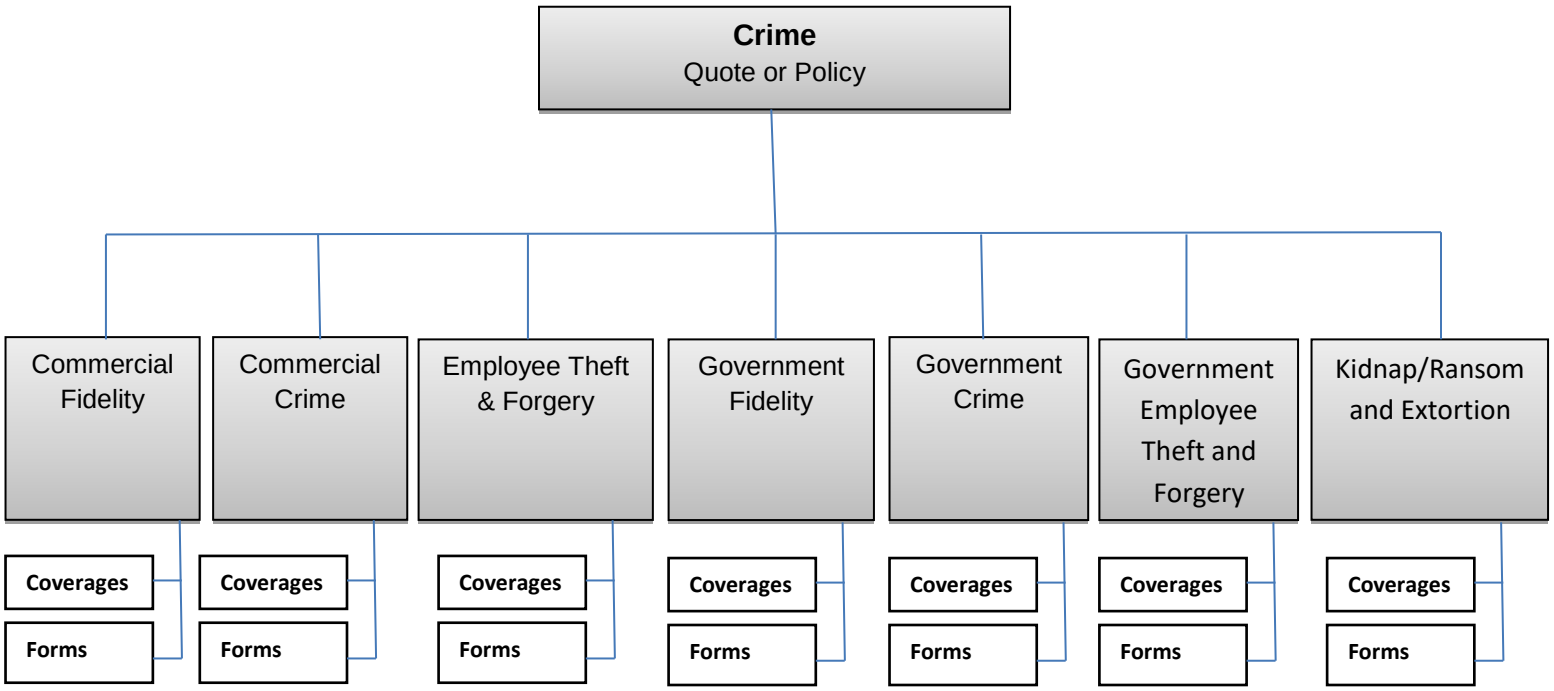
Commercial Property Hierarchy



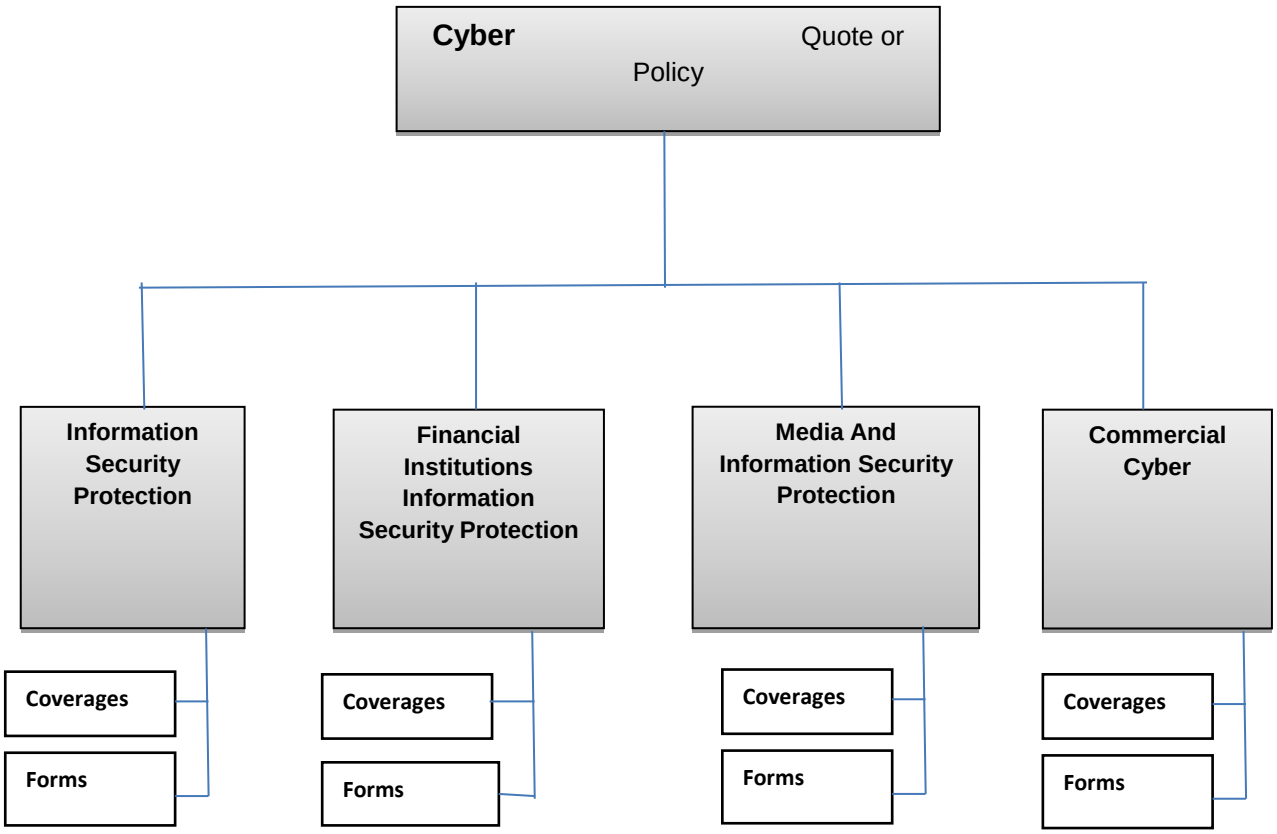
Commercial Umbrella Hierarchy



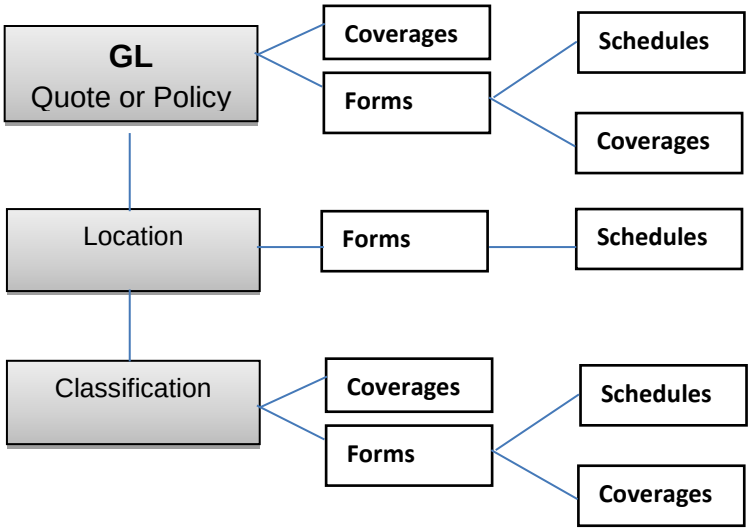
Crime Hierarchy



Cyber Hierarchy



General Liability Hierarchy



Workers Compensation Hierarchy

